

EEE 402 (January 2026)

Artificial Intelligence and Machine Learning Laboratory

Final Project Report

Section: G2 Group: 01

EEG-VJEPa: Self-Supervised JEPa for EEG Classification

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Acronyms and Notation

Acronyms and abbreviations

Acronym	Meaning
10-NN	10-nearest-neighbour agreement: the share of each point's 10 nearest neighbours in a 2-D map that carry its own label
AdamW	Adam optimiser with decoupled weight decay
AUC-PR	Area under the precision–recall curve
AUROC	Area under the receiver operating characteristic (ROC) curve
BAcc	Balanced accuracy, $\frac{1}{2} \cdot (\text{TPR} + \text{TNR})$
BatchNorm	Batch normalisation (BatchNorm1d: PyTorch's 1-D batch-normalisation layer)
BSI / pdBSI	Brain Symmetry Index / its pairwise-derived variant
BYOL	Bootstrap Your Own Latent, a momentum-encoder self-supervised method
C, FEI+C	Earlier working names for GTJ and FEI+GTJ, still used as labels in Figures 12, 17 and 19
CBraMod	Criss-Cross Brain Foundation Model, a large pre-trained EEG model
DAR / DTABR	Delta/alpha power ratio / (delta + theta)/(alpha + beta) power ratio
DUA / NEDC	Data Use Agreement / Neural Engineering Data Consortium, the TUH corpus's maintainers
EDF	European Data Format, a standard file format for EEG recordings
EEG	Electroencephalography
EEG-VJEPa / VJEPa	The published EEG adaptation of V-JEPa that this report reproduces; “VJEPa” labels its released checkpoint in some tables and figures
EMA	Exponential moving average
F1	Harmonic mean of precision and sensitivity
FEI	Frequency-masked Embedding Inference, our frequency-masking encoder
FEI+GTJ	The FEI and GTJ embeddings concatenated and read by one probe
FFT / rFFT / STFT	Fast Fourier transform / its real-input form / short-time Fourier transform
GELU / ReLU	Gaussian error linear unit / rectified linear unit (activation functions)
GRU	Gated recurrent unit
GTJ	GRU-based Time-JEPa, our spectral-dynamics encoder
I-JEPa / V-JEPa	Image / video JEPa
JEPa	Joint-Embedding Predictive Architecture
L1 / L2 / MSE	Absolute-difference (L1) distance / Euclidean (L2) length / mean squared error
LaBraM	Large Brain Model, a large pre-trained EEG model
LogReg	Logistic regression
LOSO	Leave-one-subject-out cross-validation
LRU	Linear Recurrent Unit, a long-memory recurrent layer
MLP	Multi-layer perceptron
NMT	The NMT Scalp EEG Dataset, an open clinical EEG corpus from Pakistan
PSD	Power spectral density
rand-	Prefix for a random-initialisation control: the same architecture with untrained weights
s0–s4	The five TUAB training subsets, 546 recordings each
t-SNE / UMAP	t-distributed stochastic neighbour embedding / uniform manifold approximation and projection, two 2-D visualisation methods
TPR / FPR / TNR	True-positive / false-positive / true-negative rate

Acronym	Meaning
TUAB / TUH	TUH Abnormal EEG Corpus / Temple University Hospital
ViT / ViT-M / ViT-B	Vision Transformer / the two released EEG-VJEPAs checkpoints, with 384- and 768-number tokens

Notation for FEI's objective (§3.2, Code Snippet 1)

Symbol	Meaning
x	One EEG window: 19 channels $\times$ L samples
L	Window length in samples: 1,000 (5 s at 200 Hz) for FEI pre-training; 4,000 (20 s) for GTJ's pre-training and for every FEI and GTJ probe-time embedding in Experiments 5, 6 and 8
r	FEI's frequency-removal rate for one window, drawn from $U(0, 0.7)$
M	Binary mask over the window's 501 rFFT bins (1 = removed), shared by all 19 channels
x'	The frequency-masked copy of x , $\text{irfft}(\text{rfft}(x) \cdot (1 - M))$
u, u'	Online embedding of x and EMA-target embedding of x' , after projection to h numbers
m	The mask M embedded as a prompt, $\text{mask_enc}(M)$
$\hat{u}, \hat{m}$	FEI's two predictions, of u' and of m
$n(\cdot)$	L2 normalisation, $n(v) = v/\ v\ $
$\text{sg}(\cdot)$	Stop-gradient: passes its input forward but blocks gradients flowing back through it
h	Width of the projected embeddings, 128 (the encoder output itself has 256 numbers)
$\theta_{\text{online}}, \theta_{\text{target}}, a$	Online and EMA-target weights, and the EMA momentum $a = 0.996$
ℓ	FEI's per-window pre-training loss
$\pm$	Mean $\pm$ standard deviation, over folds, training subsets or seeds as each table states

1 Abstract

EEG-VJEPa is a self-supervised model published by its authors that uses video-style JEPa pre-training for EEG, for which they report strong results on the TUH Abnormal EEG Corpus (TUAB) [1]. We wanted to recreate it and then expand upon it. The reproduction failed. The released, pre-trained checkpoints are representation-collapsed: on two completely different data sets, almost every recording we fed into them came out as nearly the same output vector, and a classifier built on top of them does no better than one built on random, untrained weights. None of the pre-processing variations we tested explained the collapse, and it seems to be a property of the released weights. After following the paper's own pipeline as far as the paper and code would allow (43 of 48 files were byte-identical to the authors' public repository) we were still unable to reach the paper's reported accuracy.

To this end, we constructed our own two-branch alternative: FEI (Frequency-masked Embedding Inference), a 1-D convolutional encoder pre-trained to predict, given an intact EEG window, how a slowly-updated copy of itself embeds the same window with random frequency bins removed; and GTJ (GRU-based Time-JEPa), a spectrogram encoder pre-trained to predict the future course of a recording's spectrum from its past. Combined (FEI+GTJ), our original pre-training run reaches an AUROC of 0.865 ± 0.024 on five-fold cross-validation on the NMT dataset (2,417 recordings). Averaged over that run and two independently seeded re-runs, it scores 0.861; this is better than a classical hand-crafted band-power baseline scored on the same folds (0.786 ± 0.009 ; better in all 5 folds, Holm-corrected $p = 0.029$). Most of that margin is from the architecture, however, as can be seen from an untrained twin of the same fused architecture, which already gives an AUROC of 0.834 (averaged over 5 random seeds). Pre-training adds +0.027 (0.861 against 0.834), better in all 5 folds but not statistically significant after correction. On a completely different hospital's data (TUAB), FEI pre-trained on NMT alone, with no TUAB recording in its pre-training (only the linear classifier on top is trained on labelled TUAB data), scored above a random-weight twin by +0.048 (95% CI [0.015, 0.108], $p = 0.0055$) AUROC, averaged over five independently seeded pre-training runs scored on the same test set, with runs, random-init seeds and test recordings resampled together (the original run alone: +0.059, $p = 0.002$). GTJ did not survive that NMT-only transfer (0.840 vs 0.863 for its random twin), so we also re-pre-trained both encoders on NMT plus the unlabelled training recordings from TUAB: the fused model then attained a result close to the released ViT-M checkpoint under our linear probe (0.862 vs 0.838, difference not significant) and beat band power (0.757), but it was not significantly better than its own random-weight twin (0.861) in any of our three pre-training runs. We then took the NMT-only encoders to a third, unrelated task: telling stroke patients apart from healthy controls in a small resting-state EEG cohort. Here, after a spectrum-matching step we designed post hoc on this same cohort (the input fixed a priori gave 0.833), FEI+GTJ scored near to the clinical Brain Symmetry Index (BSI) baseline (0.926 vs 0.907 AUROC, leave-one-subject-out: a model is validated on one subject it does not see any data from, and this is repeated once for each subject). That difference is too uncertain to be considered either superiority or equivalence, particularly with 15 subjects. Its NMT pre-training was re-run twice with seeds, scoring 0.963 and 0.981 on the same 15 subjects: a check on the pre-training seed, not on more subjects.

In this report, all pre-trained encoders are validated with a random initialisation control, with the same architecture, weights that have not been trained, and the same pipeline (one disclosed exception: the released ViT-B in Experiment 1). It is that same habit that caught the released model's collapse in the first place, and what later caught our own GTJ quietly collapsing under a spectrum shift between datasets, which we then fixed, post hoc, on the same cohort, with a step of label-free spectral matching. FEI over-flags healthy children, while GTJ is weak to distribution shift. These findings restrict where each of the encoders can be trusted.

2 Introduction

Like most clinical machine learning, automatically deciding whether a brain recording looks "normal" or "abnormal", or differentiating a stroke patient's brain activity from a healthy control's activity, runs into the same problem: labelled EEG is expensive. Each recording has to be read and annotated by a trained neurologist, so the amount of data available for supervised learning remains small compared to that used for images or text. The standard remedy is self-supervised pre-training. Train a big encoder on a task which does not require any labels, then use the trained encoder on the small labelled task you are interested in.

Meta AI introduced I-JEPa in 2023 [2] and V-JEPa in 2024 [3]. Instead of reconstructing missing pixels, the

model predicts the embedding of the missing region from the embedding of the visible region, while the target is a slowly-updated (“momentum”) copy of the model. The masking induces the learning. If you mask enough regions, the only way the context encoder can continue to be successful at this prediction is to learn how the signal's parts fit together. Raw pixel values (or raw voltage samples, for EEG) contain a wealth of detail that an encoder would waste capacity reproducing exactly. Predicting in the embedding space lets it skip that detail and keep what is actually predictable. This tends to be what a downstream classifier needs anyway. It also avoids a known pitfall of predictive self-supervision: representation collapse, which is when the easiest way to make the loss smaller is to output the same vector for everything. Keeping the two networks from sharing their weights directly is what prevents that.

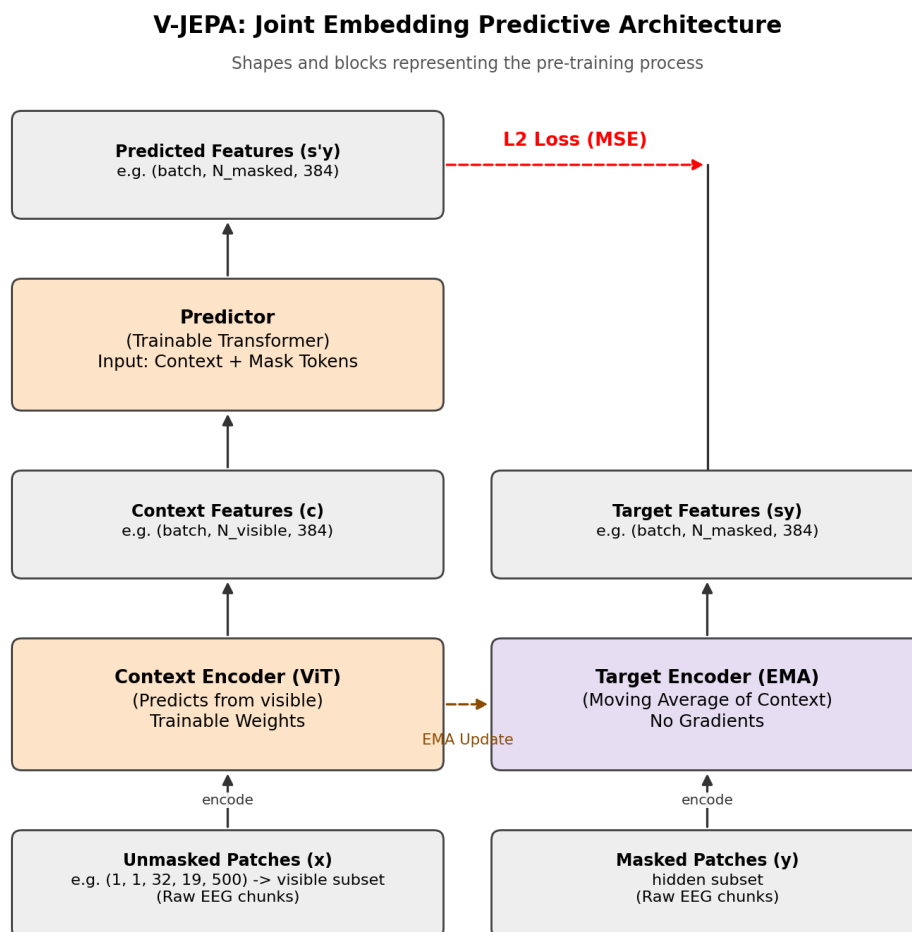


Figure 1: The JEPA idea: predict the target encoder's embedding of a masked/future region from the embedding of the context encoder, never the raw pixels [2, 3].

EEG-VJEPA [1] extends this idea to EEG by treating a multi-channel recording as a short silent video, with channels representing spatial rows and time windows representing video frames. The authors claim that their model trained in this manner and tested on TUAB achieves an AUROC of approximately 0.877–0.879, ahead of earlier published baselines. They also shared their pre-trained checkpoints and code, so choosing this paper seemed like a safe bet: replicate a recent, respectable looking accomplishment, understand its success, and then push it further on a harder or different task. The assumption that it was safe turned out to be the wrong one.

As faithfully as possible we attempted to replicate the paper's own TUAB result, using the same released weights as the paper, and the same evaluation code as the paper. That failed, and it was discovered that this failure is a condition that can be diagnosed: Released checkpoints are collapsed. This was used as the starting point for the rest of the project (“Act 0”, in our naming, since the rest of the course builds on it); we designed our own pre-trained encoder – FEI+GTJ – validated it on the dataset it was trained on (NMT), tested it on

TUAB, a different hospital's data, both with encoders pre-trained on NMT only and after we added TUAB's unlabelled recordings to pre-training; and finally applied it to a third, out-of-domain task, separating stroke patients from healthy controls, for which a well-established clinical baseline (the Brain Symmetry Index) let us check if our features add anything over classical signal processing.

3 Design

3.1 Problem Formulation

3.1.1 Identification of Scope

We initially submitted a proposal in week 5 consisting of two parts. Part 1: Reproduce EEG-VJEPa's published TUAB result for the author's published checkpoints and explain any gap. Part 2: use the same frozen encoders and a classical clinical feature (the Brain Symmetry Index) on a different cohort, on a problem of healthy vs stroke. A model that performs well on a single benchmark with which it was tuned is not very useful.

A third item was in the original proposal too: pre-training our own copy of EEG-VJEPa from scratch. We dropped it mid-term. By that time it was obvious that our remaining weeks and our one GPU would be better spent getting to the bottom of why the released model collapses—and creating a replacement that doesn't—than on a from-scratch run.

This scope is internally referred to as “Act 0” and includes reproduction, diagnosis and first working replacement. It doesn't include the motor-imagery work the team has just begun on a different task, which is out of scope here and is only reflected in Future Work.

3.1.2 Literature Review

This is anchored at one corner by the JEPa family: I-JEPa [2] and its video extension V-JEPa [3] predict embeddings rather than pixels, following in the footsteps of BYOL's momentum-encoder idea [4]: a slowly-updated target network, no negative pairs, in principle no collapse. Even that safeguard can be turned off elsewhere in a training config, as shown below in Experiment 3.

A much larger body of work is the EEG foundation models. In addition to EEG-VJEPa, two larger EEG transformers were released around the same time: CBraMod [5] and LaBraM [6] were both pre-trained using tens of thousands of hours of pooled EEG data. CBraMod: about 0.83 balanced accuracy / 0.92 AUROC on TUAB; LaBraM: about 0.81 / 0.90. These set the current bar, and gave us a sense of how a large, well-resourced EEG model is normally reported.

Our source, which we pre-train on and cross-validate against, is an open source from a South-Asian population, called NMT [7]. The EEG-VJEPa paper is focused on the U.S. benchmark TUAB [8]. The Brain Symmetry Index [9, 10] is a decades-old hand-crafted feature that measures the asymmetry of the powers of left and right, and is known to track stroke severity. We did not choose it because it is weak; it's a real yardstick for the stroke experiments. One more reference we almost missed: a methodological preprint on stress-testing EEG foundation models with negative controls (arXiv:2607.24519 [11]) appeared partway through the project, well after we'd embarked on our comparisons between all the pre-trained models and a random-init twin on the same pipeline. It proved to be an argument for the very habit we had already embraced: welcome confirmation, but not a change in approach.

3.1.3 Formulation of Problem

More specifically, the project pursues a series of questions with each one building on the answer of the previous one. Do the released EEG-VJEPa checkpoints reproduce the paper's TUAB result, under a faithful re-implementation of the authors' own evaluation protocol, and if not, why not? If they don't, can we devise and train a self-supervised encoder, using only our compute budget, that actually performs better than a hand-crafted classical baseline on the data it was trained on, and actually carries that benefit, not just its architecture, to a recording from another hospital? If this holds, does the same, frozen, unchanged encoder add anything when applied to a third task with a different structure (stroke vs. healthy), measured directly against an established clinical feature?

3.1.4 Analysis

Each of those questions above corresponds to a check listed in full in §5.3.1: question (1) against a random, untrained control (which is exactly what caught the collapse in Experiment 1); question (2) with paired statistics, and not just a single point estimate, because our folds, cohorts and seeds are all small enough that a raw accuracy number could just be noise; question (3) against the Brain Symmetry Index, which is a bar set by classical neuroscience, not by us.

3.2 Design Method

The released EEG-VJEPa [1] treats a 19-channel, 5-second EEG window as a single-frame “video clip”: channels take the place of one spatial axis and short time-slices the other, and a Vision Transformer (ViT-M or ViT-B) patch-embeds this grid just like a video frame. In pre-training, V-JEPa masks out spatiotemporal blocks of this grid and teaches a context encoder to predict a momentum target encoder's embedding of the masked blocks.

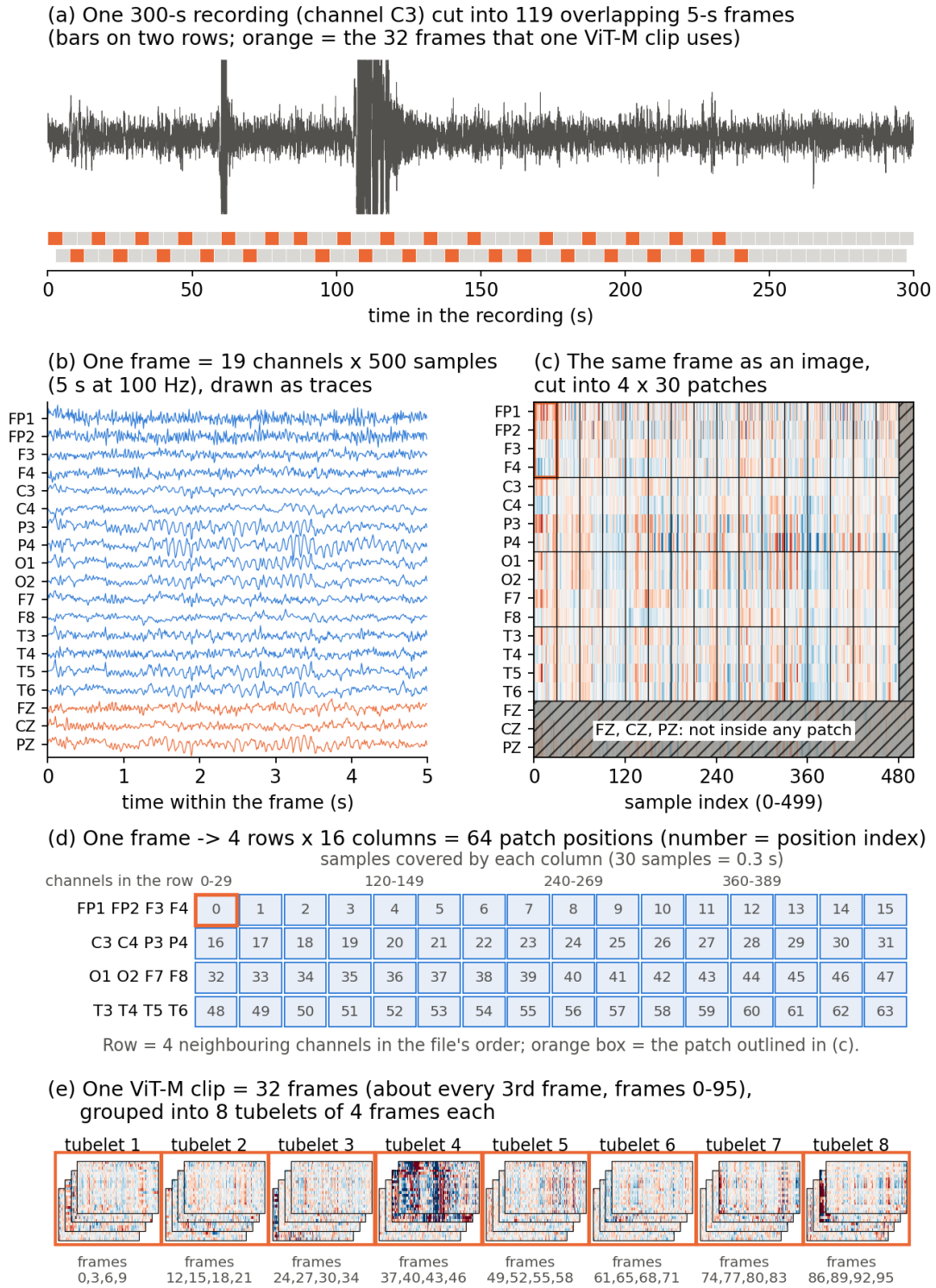


Figure 2: The released EEG-VJEPa algorithm takes a multi-channel EEG window as a single video frame to be processed by the Vision Transformer encoder [1].

Concretely, each 19×500 frame is divided into patches of $4 \text{ channels} \times 30 \text{ samples}$, and groups of 4 consecutive frames are merged together into one “tubelet” (as ordinary V-JEPa [3] does for short video clips to create a manageable number of tokens). This yields 512 tokens per clip, each described by 384 numbers for ViT-M; these numbers are then passed through 12 transformer blocks, each with 6 self-attention heads, so every token can draw on every other token, and a small MLP of $384 \rightarrow 1536 \rightarrow 384$ per block (ViT-B instead has 768-number tokens, 12 heads, and 16-frame clips).

Released ViT-M (4x30x4) encoder: shapes recorded on one real TUAB clip

Shapes are (batch, ...). Batch = 1 clip here.

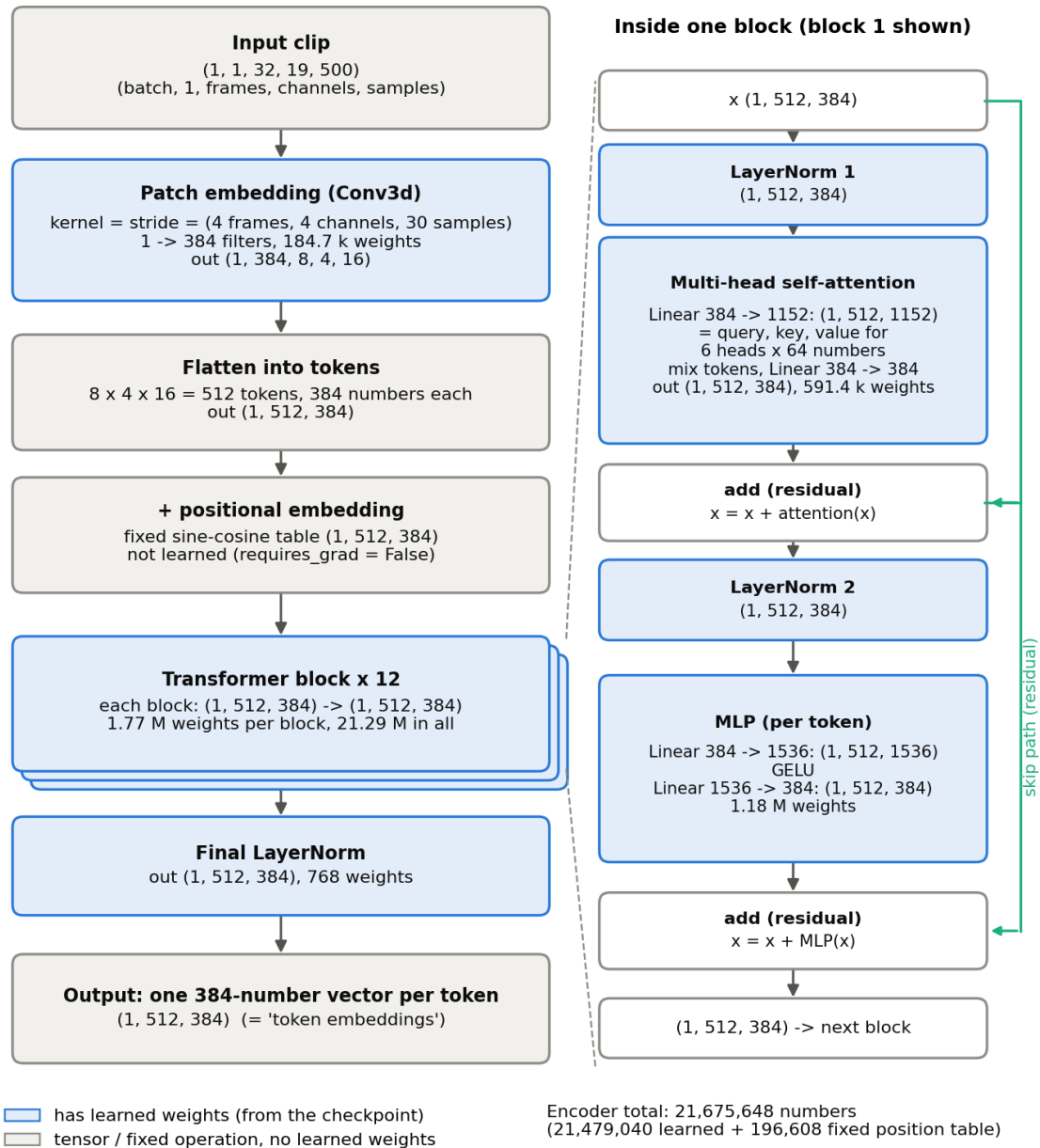


Figure 3: The Vision Transformer of EEG-VJEPa: patch and tubelet size, the resulting 512-token $\times$ 384-number grid, and the stack of self-attention blocks that process this grid [1].

This report repeatedly refers back to the three-part JEPa structure, in the released model's form: a context encoder, which is only ever presented with the visible tokens of a clip; a target encoder, which is an exponential-moving-average (EMA) copy of the context encoder, never directly trained, presented with the entire, unmasked clip, whose outputs at the masked positions become the prediction targets; and a small predictor network that is presented with the context encoder's tokens, plus the learned “mask tokens” marking the hidden positions, and tries to predict the target encoder's output at the masked positions. Only the embedding is predicted, never the raw signal. The training loss of the released checkpoint is the mean L1 distance between predicted and target embeddings, and we directly return to this in Experiment 3. Both of our own encoders below have the same three parts (online, EMA target, predictor), but what each encoder sees is different: FEI has the online encoder looking at the full window, while the target encoder is looking at a frequency-masked copy of the window; and GTJ has the online encoder looking at the window's past, and the target encoder looking at its future.

More specifically, the weights of the target encoder never get touched by gradient descent at all: after each training step the target weights are updated by a small proportion towards the current weights of the context (“online”) encoder [1], i.e., $\theta_{\text{target}} \leftarrow a \cdot \theta_{\text{target}} + (1 - a) \cdot \theta_{\text{online}}$, with momentum $a = 0.996$ as used throughout this report (thus, each update moves the target only 0.4% of the way towards the online encoder). This helps to maintain a relatively constant prediction target from the previous step to the present step, which is crucial to the predictor “catching” a slowly moving prediction target as opposed to the one that is moving as fast as the network generating it.

Real masks from the authors' mask generator (ViT-M, $8 \times 4 \times 16 = 512$ tokens)

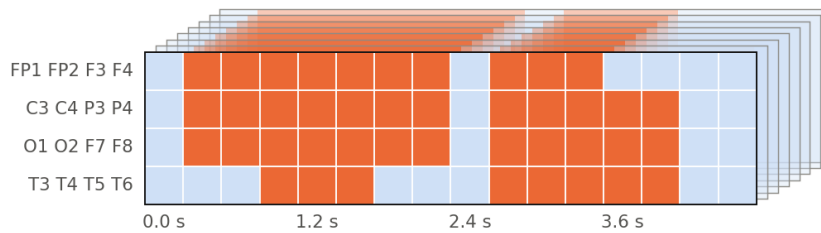
Blue = context: the encoder sees these tokens.

Orange = hidden: the predictor must guess the target encoder's vectors for these tokens.

(a) Short-range mask: 8 blocks, each asked to cover 15 % of the 4×16 slice

-> each block is 3-4 rows x 2-3 columns and spans all 8 tubelet times (the stack behind the front slice).

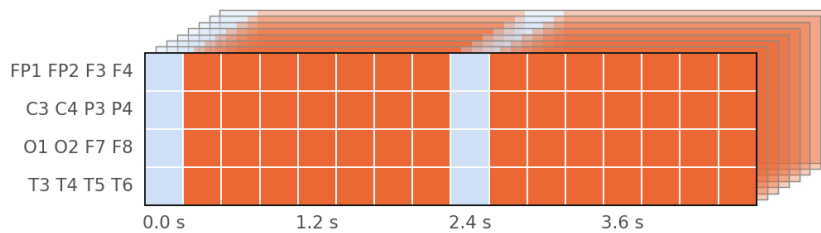
This step: context = 176 tokens, hidden = 336 of 512 (66 %).



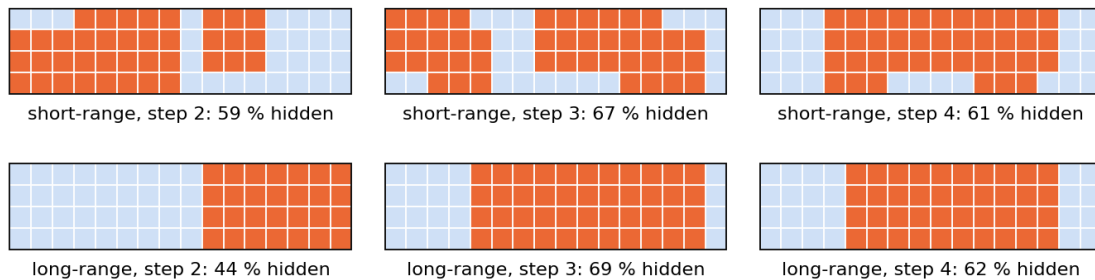
(b) Long-range mask: 2 blocks, each asked to cover 70 % of the 4×16 slice

-> each block is all 4 rows x 5-8 columns and spans all 8 tubelet times (the stack behind the front slice).

This step: context = 64 tokens, hidden = 448 of 512 (88 %).



(c) Three more training steps: a new random mask every step (front slice only).



Average over 500 steps: 68 % hidden (short-range), 60 % (long-range). Blocks can overlap, so the share varies.

Figure 4: The context encoder is only presented with the unshaded tokens and the target encoder is presented with all the tokens and provides the target tokens that need to be predicted at the shaded, masked positions [1].

The released checkpoint was collapsed and unrescuable by fine-tuning (Experiments 1–4, Results section), so we designed our own encoder pair from scratch around two principles that were suggested by the failure of the released model: mask in the frequency domain, where pathological EEG slowing can be directly observed as a spectral shift that a time-domain patch mask can miss; and keep the architecture small and match it to the size of the dataset (NMT has only a few thousand recordings, far fewer than the millions of video clips typical JEPA training uses), so that the encoder is not so over-parameterised that ignoring the input becomes the easiest way to solve its pretext task.

FEI is a 1-D convolutional Encoder. Its pretext task removes a random fraction (between 0% and 70% of rFFT frequency bins, redrawn independently for each window, $\mathcal{U}(0, 0.7)$ in code/fei_pretrain.py) of the FFT frequency content of the signal and uses an online encoder to predict, in embedding space, what a slowly-

updated momentum copy of itself sees when given the masked window, plus a minor auxiliary term that predicts which bands were removed. Both predictions and targets are L2-normalized before the loss is computed (a BYOL-style trick that is the same as a bounded $2-2\cdot\cos(\text{pred}, \text{target})$ loss), making a trivial solution to the problem – namely a network that shrinks its output scale to zero to get the loss to be 0, which an un-normalised MSE loss would allow – impossible.

```
class FEI(nn.Module):
    def __init__(self, n_freq, d=256, h=128, in_ch=NCH):
        super().__init__()
        self.enc, self.proj = Encoder(d, ch=in_ch), nn.Linear(d, h) # online
        self.enc_t, self.proj_t = Encoder(d, ch=in_ch), nn.Linear(d, h) # momentum target
        self._sync(); [p.requires_grad_(False) for p in self._tgt_params()]
        self.mask_enc = nn.Linear(n_freq, h, bias=False) # mask prompt -> embedding space
        self.z1, self.z2 = mlp(h), mlp(h) # predictors

    def forward(self, x, xp, M):
        e = self.enc(x); u = self.proj(e) # online, original window
        with torch.no_grad():
            up = self.proj_t(self.enc_t(xp)) # momentum, masked window
            m = self.mask_enc(M)
            uhat = self.z1(u + m.detach()) # predict masked embedding
            mhat = self.z2(u.detach() - up) # predict which bands were
removed
            nrm = lambda a: F.normalize(a, dim=-1) # BYOL-style: bounded loss, no
scale collapse
            l_embed = F.mse_loss(nrm(uhat), nrm(up), reduction="none").mean(-1)
            return l_embed.mean() + F.mse_loss(nrm(mhat), nrm(m)), e, l_embed.detach()

def freq_mask(x, bias=0.0, b1=0.0, b2=0.7):
    """Zero a random U(b1,b2) fraction of rFFT bins (shared across channels)."""
    Xf = torch.fft.rfft(x, dim=-1) # (B,19,n) complex
    ratio = torch.empty(x.shape[0], 1, device=x.device).uniform_(b1, b2)
    M = (torch.rand(x.shape[0], Xf.shape[-1], device=x.device) < ratio).float()
    xp = torch.fft.irfft(Xf * (1 - M).unsqueeze(1), n=x.shape[-1], dim=-1)
    return xp.float(), M
```

Code Snippet 1: FEI's momentum-encoder JEPA loss and its frequency-domain masking function (code/fei_pretrain.py, simplified – there is also an adaptive, difficulty-weighted masking variant, `--mask\_bias`, that is only touched upon in the fused scores of GTJ's window-length sweep below).

Written out once, for one window x (19 channels $\times$ L samples, $L = 1,000$ in pre-training): `freq_mask` draws a removal rate $r \sim U(0, 0.7)$, removes each of the 501 rFFT bins independently with probability r (the same bins on every channel, recorded in a binary mask M), and inverts the FFT to give the masked copy $x' = \text{irfft}(\text{rfft}(x) \cdot (1 - M))$. The online path embeds the original window, $u = \text{proj}(\text{enc}(x))$; the target path embeds the masked copy without gradient, $u' = \text{proj}_t(\text{enc}_t(x'))$; and the mask itself is embedded as a prompt, $m = \text{mask_enc}(M)$. The two predictors are $\hat{u} = z_1(u + \text{sg}(m))$ and $\hat{m} = z_2(\text{sg}(u) - u')$, where $\text{sg}(\cdot)$ is stop-gradient; with $n(v) = v/\|v\|$ and $h = 128$, the loss, averaged over the batch, is

$$\ell = (1/h) \cdot \|n(\hat{u}) - n(u')\|^2 + (1/h) \cdot \|n(\hat{m}) - n(m)\|^2$$

The first term trains the encoder, its projection and z_1 ; the second term trains only z_2 and the mask-prompt encoder, because $\text{sg}(\cdot)$ stops its gradient from reaching u . The target weights (`enc_t`, `proj_t`) are never trained by gradient; after each step, they are pulled towards the online weights by the EMA rule above, with $\alpha = 0.996$.

That whole step is done in one shot in the figure below, for a particular batch of 4 windows: the frequency masking, the online and target encoders, both predictor heads, and where the EMA update occurs.

One FEI pre-training step (real shapes, batch of 4 windows)

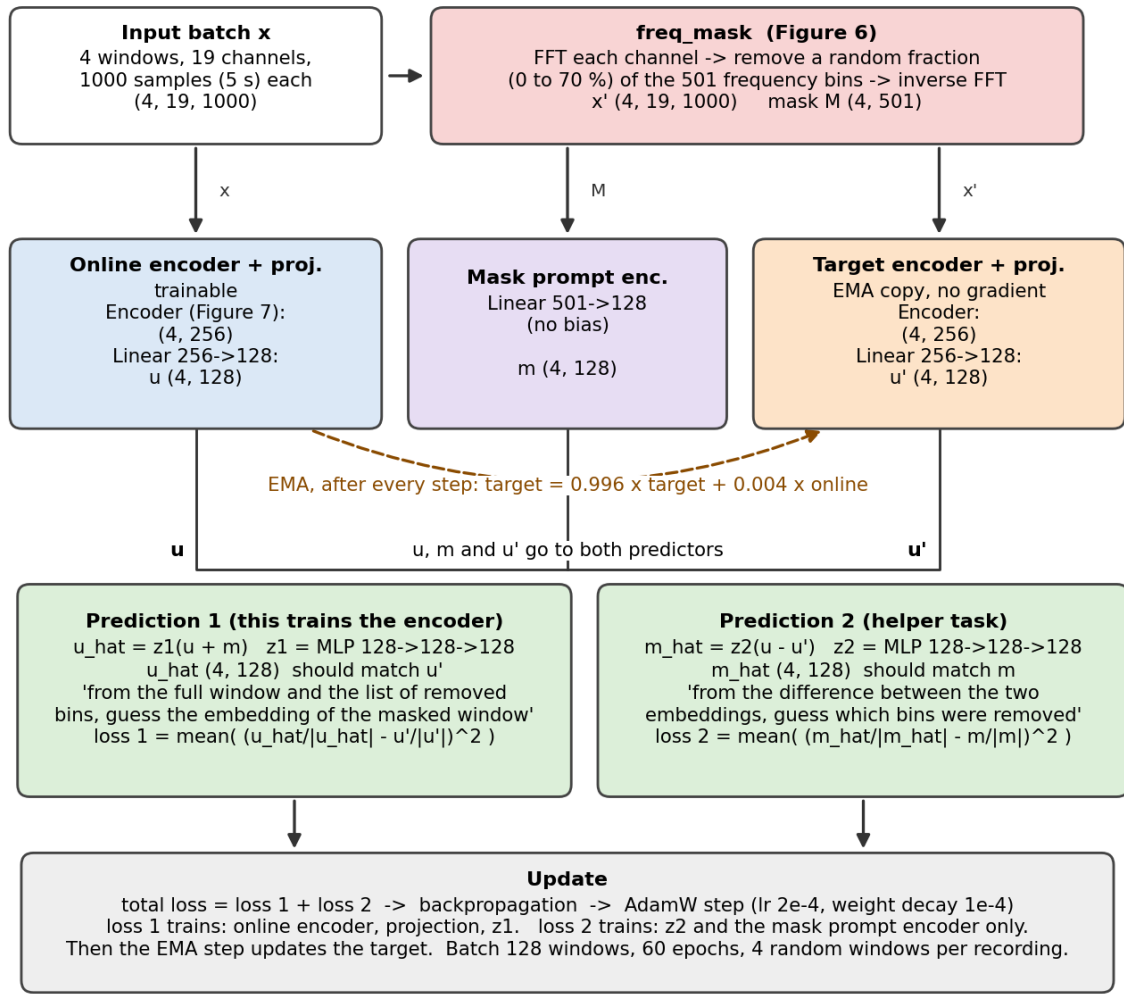


Figure 5: One FEI pre-training step for a batch of 4 windows: frequency masking produces the masked copy x' and the mask M , and the masked copy x' is passed to the EMA target encoder while the online encoder embeds the original window x ; prediction 1 recovers the target's masked-window embedding from the online embedding plus the mask prompt, and prediction 2 recovers the mask prompt from the difference between the two embeddings; the online branch and the mask-prompt encoder are updated by AdamW and the target branch is updated by EMA.

The next figure shows freq_mask operating on one real NMT channel's window instead of the abstract: which rFFT bins are zeroed in one draw and what effect that zeroing has on the time-domain signal.

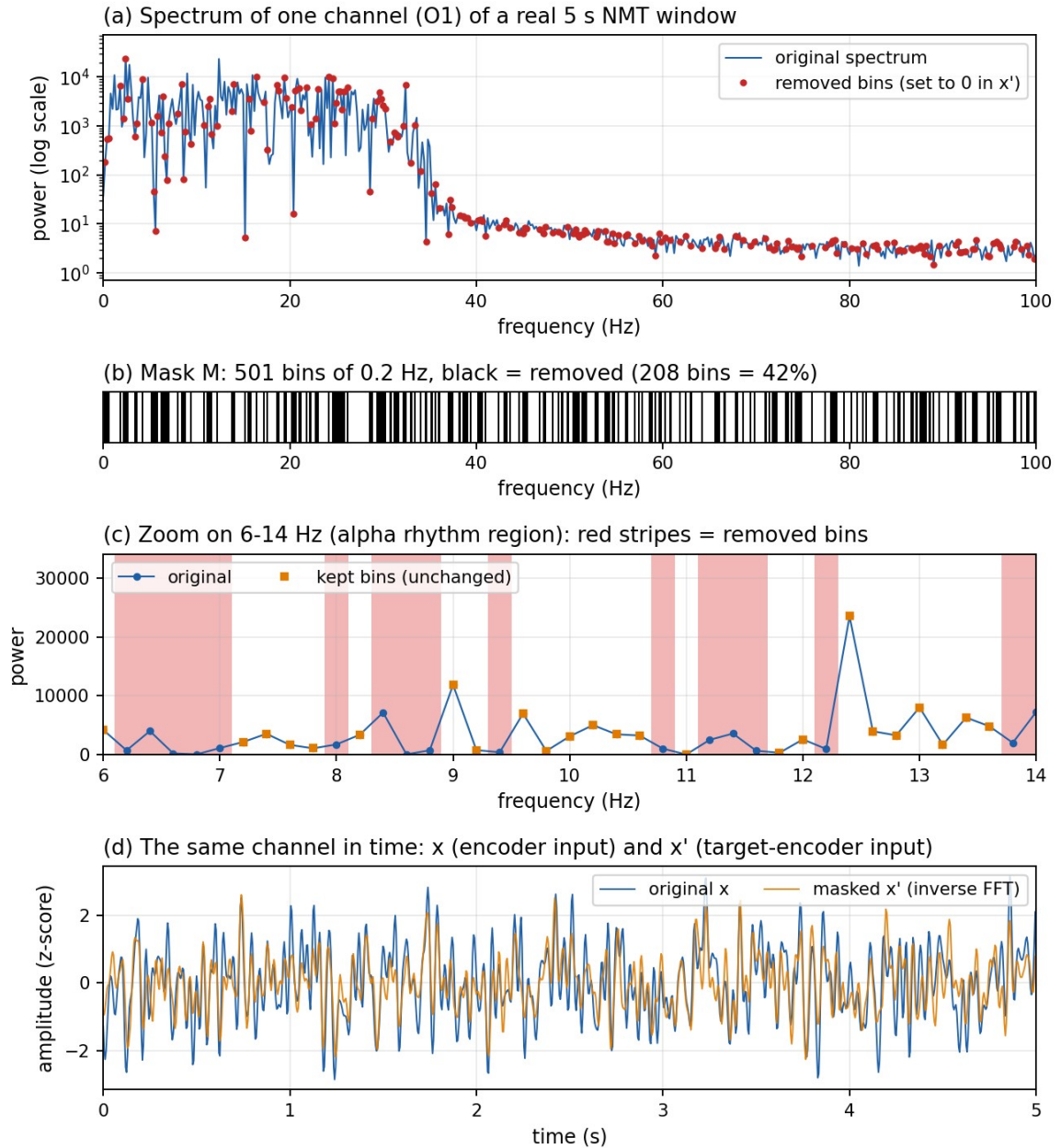
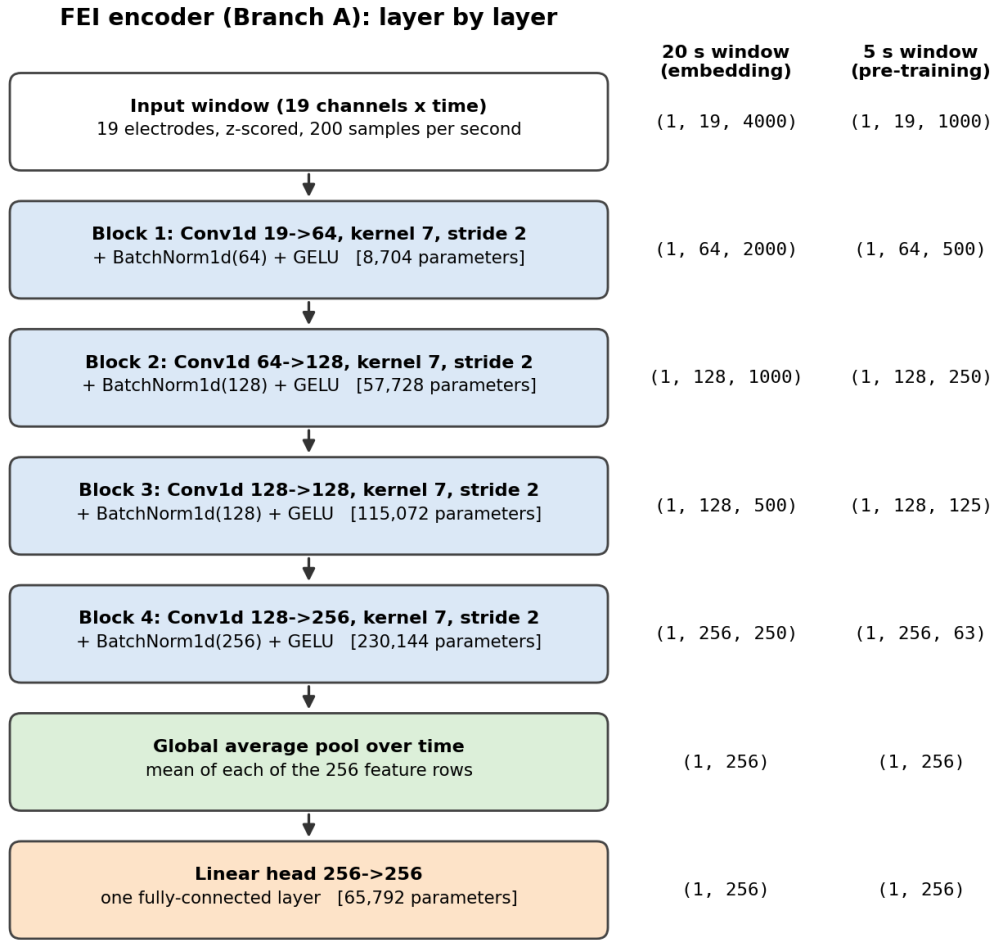


Figure 6: freq_mask applied to one real NMT channel's 5-second window: the original spectrum, with the bins a real draw removed marked in red (a), the binary mask itself, with 208 of 501 bins (42%) removed in this draw (b), a zoom of the alpha band (6-14 Hz) where it can be seen exactly which bins survive (c), and the resulting time-domain signal before and after applying the mask and inverse-FFT (d).

In structure, the `Encoder` in FEI is a small 1-D convolutional network, not a transformer: four blocks, each a Conv1d (kernel 7, stride 2) → BatchNorm → GELU, where the number of channels goes 19 → 64 → 128 → 128 → 256 (19 input channels, 256 output channels); stride 2 reduces the length of time by half at each block; an average over the remaining time steps yields 256 numbers; and a final Linear layer yields the 256-number embedding used throughout this report.



Output: one embedding of 256 numbers per window, whatever the window length.
 Shapes are (batch, channels or features, time steps), recorded by forward hooks on
 fei_pretrain.Encoder with checkpoint fei_enc_cv_s0_f0.pt. Each block halves the time axis;
 one step after block 4 sees 91 input samples = 0.46 s. Encoder: 477,440 parameters.

Figure 7: We use FEI's convolutional encoder, which is made of 4 Conv1d/BatchNorm/GELU blocks that reduce 19 raw EEG channels to a 256-number embedding.

Table 1: FEI, layer by layer, with exact parameter counts instantiated from code/fei_pretrain.py at the actual Act-0 config: L = 1000 samples = 5 s at 200 Hz, n_freq = 501, d = 256, h = 128, 19 channels. The target branch is not trained directly, but instead is an EMA copy of the weights from the online branch, so it has exactly the same number of parameters as the online encoder.

Component	Layer	Shape (in → out)	Parameters
Encoder (online + target, shared structure)	Conv1d block 1 + BatchNorm1d	19 → 64, k=7, s=2	8,704
	Conv1d block 2 + BatchNorm1d	64 → 128, k=7, s=2	57,728
	Conv1d block 3 + BatchNorm1d	128 → 128, k=7, s=2	115,072
	Conv1d block 4 + BatchNorm1d	128 → 256, k=7, s=2	230,144
	Global avg-pool + head Linear	256 → 256	65,792
	proj Linear	256 → 128 (h)	32,896

Component	Layer	Shape (in → out)	Parameters
mask_enc	Linear (mask prompt → embedding space)	501 (n_freq) → 128 (h)	64,128
z1 predictor	Linear → GELU → Linear	128 → 128 → 128	33,024
z2 predictor	Linear → GELU → Linear	128 → 128 → 128	33,024
Online branch total	enc + proj (trainable)	—	510,336
Target branch total	enc_t + proj_t (EMA, frozen)	—	510,336
TOTAL	online + target + mask_enc + z1 + z2	—	1,150,848 (1.15M)
TRAINABLE subset	everything except enc_t, proj_t	—	640,512 (0.64M)

FEI is trained with AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}), batch size 128, for 60 epochs with 4 random 5-second windows per recording per epoch (`--wpe` in code/fei_pretrain.py), instead of always the same crop. At probe time, the single 256-number embedding of a recording (`embed\_recording`) is the output mean of the online encoder over every non-overlapping window in that recording, rather than the embedding of one arbitrary window. The encoder ends in a global average over time, so any window size can be used, and every FEI number in Experiments 5, 6 and 8 uses 20-second windows (4,000 samples), the same size used in GTJ, even though FEI is pre-trained on 5-second windows.

GTJ (GRU-based Time-JEPA) attacks a different vulnerability: a hand-crafted band-power feature averages the power of the spectrum over a whole window, discarding the information about how the spectrum changes over time. GTJ instead maintains a short-time Fourier transform (STFT) spectrogram (frequency $\times$ time) and trains an encoder based on a GRU to predict the embedding of the future part of a recording's spectrogram from the embedding of the past part. The context encoder is never shown the future frames, so this pretext task cannot be trivially shortcut, as an earlier masked-input design we attempted could (see Novelty, below).

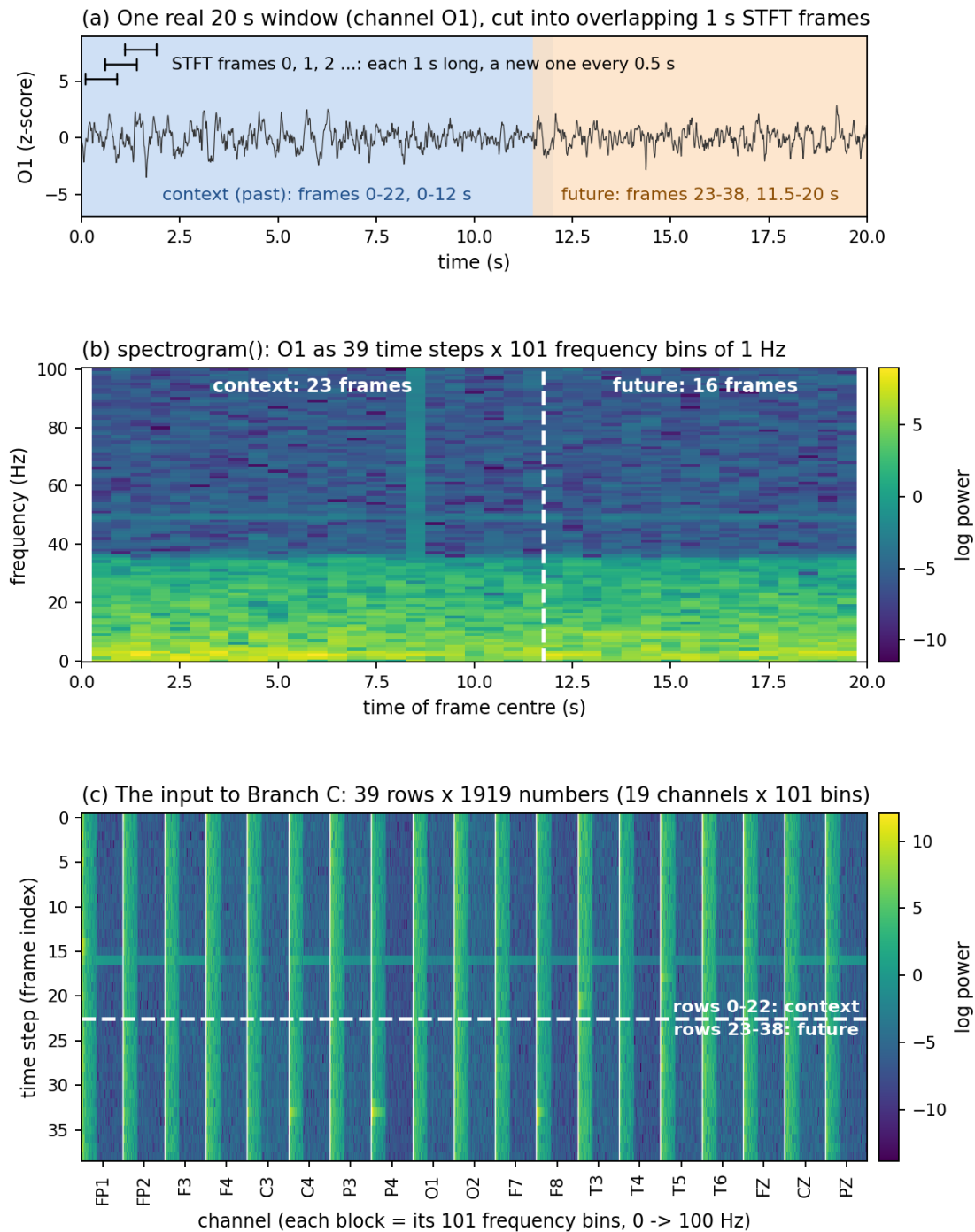


Figure 8: Each EEG window is converted to a log-power spectrogram (frequency $\times$ time) and a GRU encoder is trained to predict the embedding of the future span given the embedding of the past span.

```
def spectrogram(x):
    """(B,19,L) -> (B, T, 19*NF) log-power frames: a temporal SEQUENCE of spectra."""
    z = torch.stft(x.reshape(B_*C, L), n_fft=NFFT, hop_length=HOP, win_length=NFFT,
                  window=_WIN.to(x.device), center=False, return_complex=True)
    p = (z.real ** 2 + z.imag ** 2 + 1e-6).log() # log-power
    return p.reshape(B_, C, NF, -1).permute(0, 3, 1, 2).reshape(B_, -1, C * NF)

class DynEncoder(nn.Module):
    """Per-frame spectral embed + GRU over time -> summary of a frame span."""
    def __init__(self, d=256, in_dim=None):
        super().__init__()
        self.bn = nn.BatchNorm1d(in_dim); self.embed = nn.Linear(in_dim, d)
        self.gru = nn.GRU(d, d, batch_first=True)
```

```

    def forward(self, z):
        # z: (B,T,in_dim)
        h = torch.relu(self.embed(self.bn(z.reshape(-1, z.shape[-1])).reshape(z.shape)))
        out, _ = self.gru(h)
        return out[:, -1]
        # last hidden = summary of
the span

class DynJEPA(nn.Module):
    """Predict FUTURE-span embedding from PAST-span embedding. BYOL-style, EMA target."""
    def forward(self, z):
        past, future = z[:, :TC], z[:, TC:]
        u = self.proj(self.enc(past))
        with torch.no_grad():
            v = self.proj_t(self.enc_t(future))
        nrm = lambda a: F.normalize(a, dim=-1)
        return F.mse_loss(nrm(self.pred(u)), nrm(v))

```

Code Snippet 2: GTJ's spectrogram construction and its past-predicts-future JEPA objective (code/fei_branchC.py, simplified).

Here, the same idea is shown as a picture, for one concrete batch: the spectrogram is split into past and future spans, each summarized by the shared DynEncoder, and the output of the predictor on the past span is pushed towards the future-span embedding of the EMA target.

GTJ: Spectral-Dynamics JEPA

Predicts the future-span embedding from the past-span context

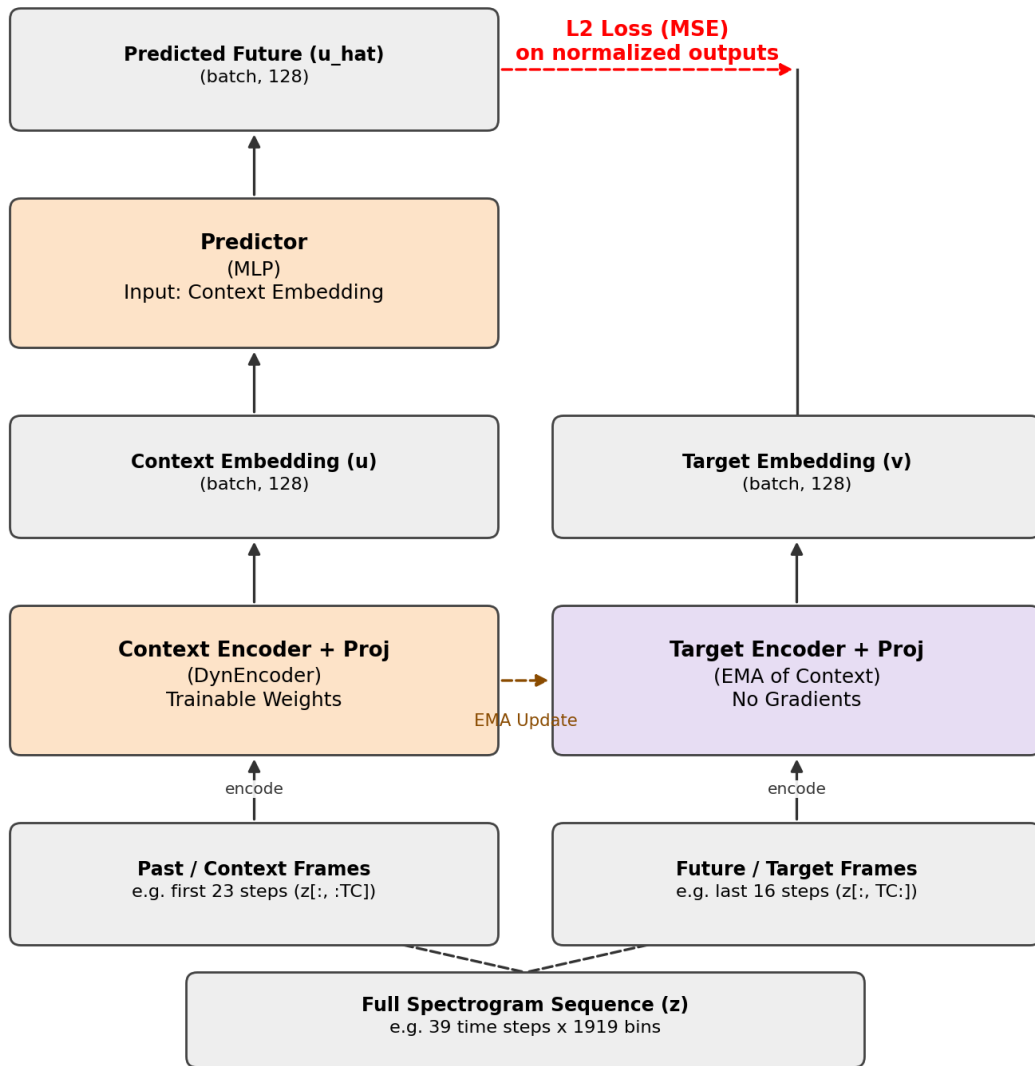
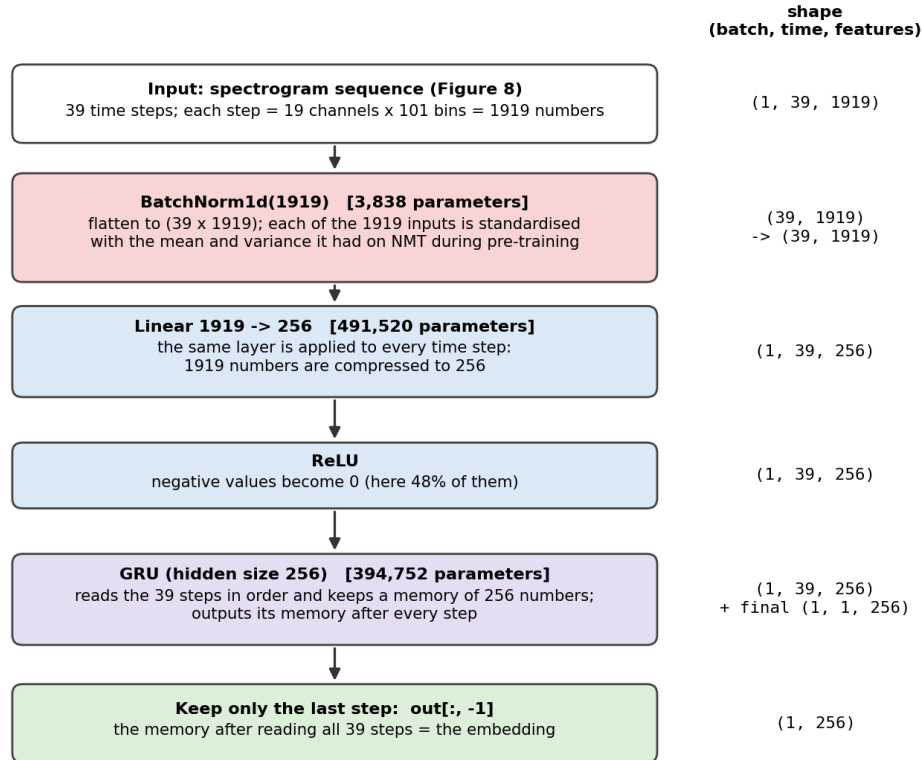


Figure 9: First 23 frames (past/context) and last 16 frames (future/target) of the spectrogram are each represented by the shared DynEncoder structure (online weights for the past span, and the EMA target copy for the future span); the predictor MLP maps the past embedding towards the future embedding, and the normalized squared error between them is backpropagated into the online encoder, with the target encoder following by EMA, just like in FEL.

The following figure divides the DynEncoder itself into a sequence of layers, and also shows a real example's GRU hidden-unit activations on all 39 frames of one recording's spectrogram.

GTJ encoder (DynEncoder): layer by layer



BatchNorm also stores 3,838 running statistics (mean and variance of each input). They are not trained by gradients; in eval mode they are frozen at their NMT values. Encoder: 890,110 parameters. Shapes recorded by forward hooks on fei_branchC.DynEncoder, checkpoint branchC_h20_enc_cv_s0_f0.pt.

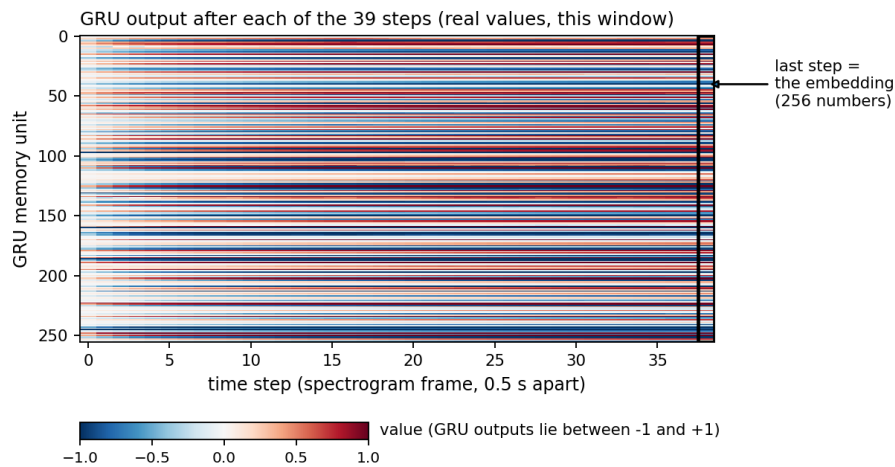


Figure 10: GTJ's DynEncoder, layer by layer — BatchNorm1d over a 1,919-dim input, a Linear embedding down to 256 dimensions with a ReLU, and a 256-unit GRU (only the last hidden state is kept) — and a real heatmap of that GRU's hidden-unit activations across all 39 spectrogram frames for one window.

Table 2: The GTJ, layer by layer, with exact parameter counts instantiated from code/fei_branchC.py at real Act-0 config (L=4,000 samples = 20 s at 200 Hz, nfft=200, hop=100, giving NT=39 frames, NF=101 freq bins, 23 frame past / 16 frame future split). The number of parameters in the target branch is exactly the same as the online branch since it is an EMA copy and not an independently trained network, like in FEL.

Component	Layer	Shape (in → out)	Parameters
Input (per window)	Log-power STFT spectrogram	19 ch × 101 freq bins × 39 frames → 1,919-dim/frame	—
DynEncoder (online + target, shared structure)	BatchNorm1d	1,919	3,838

Component	Layer	Shape (in → out)	Parameters
	Linear (embed) + ReLU	1,919 → 256	491,520
	GRU (1 layer, last hidden state)	256 → 256	394,752
	proj Linear	256 → 128 (h)	32,896
pred predictor	Linear → GELU → Linear	128 → 128 → 128	33,024
Online branch total	enc + proj (trainable)	—	923,006
Target branch total	enc_t + proj_t (EMA, frozen)	—	923,006
TOTAL	online + target + pred	—	1,879,036 (1.88M)
TRAINABLE subset	everything except enc_t, proj_t	—	956,030 (0.96M)

GTJ shares FEI's optimizer and schedule exactly (AdamW, learning rate 2×10^{-4} , weight decay 10^{-4} , batch size 128, 60 epochs, 4 random windows per recording per epoch), and the embedding of a recording at probe time is likewise the output of its online encoder, mean-pooled over each non-overlapping window of 20 seconds (`embed_dyn` in `code/fei_branchC.py`). Table 2's window length and recurrent unit are not an arbitrary choice; they were chosen as part of a dose-response test. When everything else is fixed and the window is swept from 5 s to 20 s to 40 s, GTJ alone achieves 0.820, 0.836 and 0.823 AUROC, and the difference between the learned embedding and a random-init encoder with the same architecture monotonically increases with window length — +0.021 (4/5 folds), +0.038 (5/5), +0.045 (5/5) — suggesting the embedding was learned using the temporal dynamics, not just the architecture. After replacing the GRU with a long-memory Linear Recurrent Unit (LRU) [12] (`LRUEncoder` in `code/fei_branchC.py`, a diagonal recurrence $h_t = \lambda \odot h_{t-1} + \gamma \odot (Bx_t)$ with λ initialised close to 1 for long memory) at 40 s, the performance of GTJ was basically unchanged compared to the GRU (0.819 ± 0.010 vs 0.823 ± 0.009). So the 40 s dip is because a longer, more stationary window has less to predict, and not because the GRU is running out of memory, and the cheaper GRU at 20 s is this report's GTJ configuration throughout. The fused scores on this sweep (0.858, 0.870 and 0.863, and 0.861 with the LRU) pair GTJ with the adaptive-masking FEI variant noted under Code Snippet 1, not the plain FEI used for the rest of this report, which is why the 20-second figure (0.870 ± 0.023) differs from Table 11's 0.865. There is one caveat here: the 20-second window was selected on these same seed-0 folds, and there was no nested inner split at all, so GTJ's 20-second numbers carry some selection optimism. They do not depend on it, though: GTJ beats Table 11's band power (0.786) at every window length in the sweep (0.820, 0.836 and 0.823).

Each branch is frozen after pre-training and the 256-dimensional embeddings from both branches are concatenated (FEI+GTJ; 512-dimensional) before a standard StandardScaler + class-balanced logistic-regression probe is trained on top of it (the same lightweight, low-capacity classifier used throughout this report for each model), eliminating the possibility of comparing models that have received a different level of classifier from each other.

3.3 Data and Preprocessing Pipeline

There are three datasets that carry this report, each with a unique role and each prepared using its own matched pipeline, to ensure comparisons are made fairly.

Table 3: The three data sets used in Act 0 and why they were used.

Dataset	Origin / size	Task & role in Act 0	Notes
NMT Scalp EEG [7]	Pakistan (NUST/MH); 2,417 recordings, 415 abnormal	Primary normal-vs-abnormal task; pre-trains and cross-validates FEI, GTJ and FEI+GTJ	Fully open, no licensing restriction

Dataset	Origin / size	Task & role in Act 0	Notes
TUH Abnormal EEG (TUAB) [8]	USA (Temple University Hospital); 2,993 recordings (its 276-recording evaluation split used here)	Cross-hospital transfer test for the released checkpoints and for FEI+GTJ	Our copy is an unofficial Kaggle mirror — see Legal Issues (§5.5.3); used for internal evaluation only
Stroke resting-state EEG [13]	Zenodo record 19599466; 9 stroke vs 6 control (primary cohort), 18 subjects (sensitivity cohort)	Healthy-vs-stroke transfer test, benchmarked against the clinical BSI baseline	Open Zenodo release; primary cohort only reported per team decision

This report contains two preprocessing families, and Table 4 specifies which experiments use which. The released EEG-VJEPA checkpoints only accept the input that they were pre-trained on, so all recordings they encounter are processed using the paper's own pipeline (§4.1 of arXiv:2507.03633) [1], followed as closely as its text allows (the three points it leaves open are detailed below, after Code Snippet 3): pick and reorder the same 19 standard 10–20 channels, crop to a fixed 5-minute window, band-pass filter to 1–40 Hz, resample to 100 Hz, z-score each channel, and cut the signal into 5-second frames with a 50% overlap (Code Snippet 3). The TUAB band-power baseline of Experiments 4 and 6 reads the same 100 Hz frames. FEI and GTJ instead work at 200 Hz, which is the native sampling rate of NMT: on NMT they read the recordings only per-channel z-scored (no band-pass, no resampling), and stored as 2.5 s frames; on TUAB and the stroke cohort, each recording is resampled to 200 Hz and z-scored, with a band-pass of 0.5-40 Hz on TUAB, and a common average reference plus the spectral-matching step described below on stroke. We intentionally re-derived the 100 Hz pipeline from the description in the paper, not assuming the 200 Hz pipeline we had previously preprocessed would be compatible with the released checkpoints. One of the early concerns in this project, since resolved, was whether a sample-rate mismatch could explain any gap to the paper; it could not, but getting the released checkpoints' input just right was a critical first step that wasn't obvious.

The z-score step is a per channel, per recording step to standardize the raw samples of each channel: $z = (x - \mu) / \sigma$, where x is the raw samples of each channel, μ and σ are the mean and standard deviation of that channel over the cropped recording (using $\sigma = 1$ on the rare flat channel where $\sigma = 0$, to avoid a divide by zero error). Because the electrode impedance and amplifier gain change significantly between recordings, and between the two hospitals in particular, NMT and TUAB, a scale established across the entire data set is not necessarily the best here, and will result in some recordings being under or oversaturated with respect to what the encoder saw during pre-training.

```
# Paper preprocessing (arXiv:2507.03633 §4.1), as closely as its text allows:
CHANNELS = ['FP1', 'FP2', 'F3', 'F4', 'C3', 'C4', 'P3', 'P4', 'O1', 'O2',
            'F7', 'F8', 'T3', 'T4', 'T5', 'T6', 'FZ', 'CZ', 'PZ'] # this order
TARGET_FS, WINDOW_SIZE = 100, 500 # 100 Hz, 5 s frames
CROP_SECONDS = 300 # crop to a fixed 5-minute length
BANDPASS = (1.0, 40.0) # bandpass 1-40 Hz (encoder was pretrained on this band)
WINDOW_STRIDE = 250 # 50% overlap -> (30000-500)//250 + 1 = 119 frames (paper: 118)
```

```
def preprocess_edf(edf_path, out_path, crop_start=0.0):
    raw = mne.io.read_raw_edf(edf_path, preload=False, verbose=False)
    raw.pick_channels(CHANNELS); raw.reorder_channels(CHANNELS)
    t0 = min(crop_start, max(0.0, raw.times[-1] - CROP_SECONDS))
    raw.crop(tmin=t0, tmax=min(t0 + CROP_SECONDS, raw.times[-1]))
    raw.load_data(verbose=False)
    raw.filter(BANDPASS[0], BANDPASS[1], verbose=False)
    if round(raw.info['sfreq']) != TARGET_FS:
        raw.resample(TARGET_FS)
    data = raw.get_data()
    mean, std = data.mean(1, keepdims=True), data.std(1, keepdims=True)
    data = (data - mean) / np.where(std == 0, 1.0, std) # per-channel z-score
    num_frames = (data.shape[1] - WINDOW_SIZE) // WINDOW_STRIDE + 1
    frames = np.stack([data[:, i*WINDOW_STRIDE : i*WINDOW_STRIDE+WINDOW_SIZE]
                       for i in range(num_frames)]).astype(np.float32)
    np.save(out_path, frames)
```

Code Snippet 3: The paper's own §4.1 preprocessing [1], applied to every recording the released checkpoints see, as performed in code/preprocess_nmt.py (simplified; the same filtering, resampling, z-scoring and framing is also performed on the stroke cohort in code/stroke_phaseB.py after a common average reference).

There are three open points in §4.1, each addressed in Code Snippet 3. It filters at the native sampling rate and then resamples, whereas the paper lists downsampling before filtering; it z-scores each channel over the whole cropped recording, whereas the paper's "last step" could also be read as per window (both readings give the same collapsed output from the released ViT-M, Experiment 3); and it keeps all full 500-sample frames at a stride of 250, giving $(30,000 - 500) / 250 + 1 = 119$ frames per 5-minute crop as opposed to the 118 frames the paper states. All 2,993 preprocessed TUAB files have shape $119 \times 19 \times 500$. Filtering and resampling are done using MNE 1.8.0's defaults (zero-phase FIR band-pass followed by FFT-based resampling). A recording that is less than 5 minutes will be kept at its full length, and a recording that does not contain all 19 channels will not pass the channel selection and will be skipped (the script will log it and continue). The encoder that is released reads clips of 32 frames spread evenly over 96 consecutive frames, thus resulting in 23 clip positions in a 119-frame recording. It also divides each of the 19×500 frames into patches of 4 channels $\times$ 30 samples, meaning that only the first 16 channel rows and the first 480 samples are ever embedded: in the paper's channel order, the three midline channels, FZ, CZ and PZ, come last and are never sent to the released encoder.

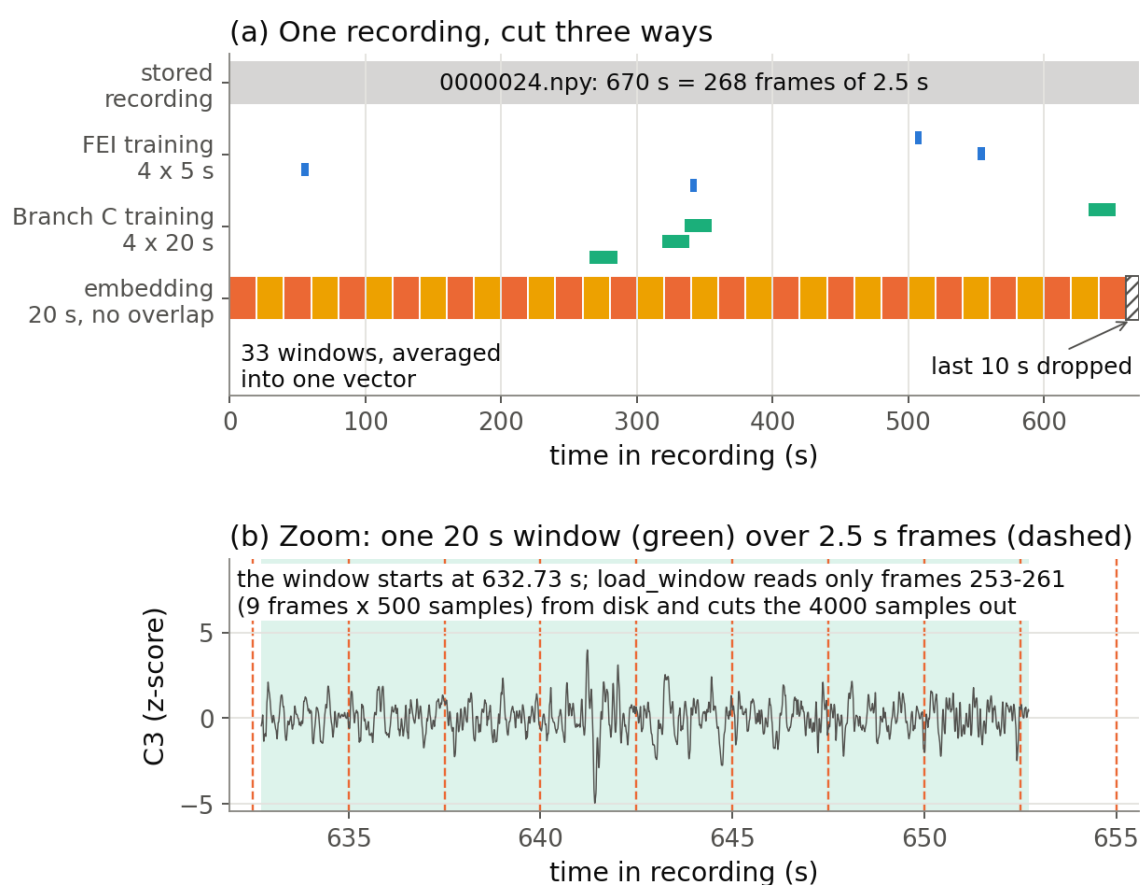


Figure 11: Segmentation of the continuous EEG recording into fixed-length windows, with 50% overlap, to feed into the encoder.

The stroke cohort required two pre-processing steps that were not required for the other two datasets. A first step of purely data quality: before any other operation is performed on the recording, any channel marked as bad by the dataset's own authors is rebuilt from the neighbouring channels, using spherical-spline interpolation. Second, a non-obvious step, fully described in §5.3.3 (Experiment 8): its recording devices and montage differ from NMT's, so the raw spectrum does not correspond to what our NMT-trained encoders expect to see, and GTJ is especially sensitive to this mismatch. We developed a mapping filter that is label-free, per-channel and matches the mean spectrum of the stroke cohort with NMT's before any encoder ever sees it, fixing this without ever looking at the stroke/control labels. The filter is fitted once on all 15 subjects pooled, so it is transductive (each held-out subject helps shape the filter applied to it); it was not pre-registered, and it was added after the pre-registered analysis had been run, for which there is a record and a fold-local

check in Appendix §11.2. Throughout, "pre-registered" indicates a pre-analysis protocol that is timestamped locally (Appendix §11.2), not an entry in a public registry.

Table 4: Which pre-processing for which experiment. Released checkpoints and TUAB band-power baseline have 100 Hz pipeline, while FEI, GTJ and the NMT band-power arm have 200 Hz. All encoders inputs are channel normalized (z-scored).

Model / experiment	Dataset	Sampling rate, filtering	Window	Samples / window	Overlap	Script
Released ViT-M/ViT-B (Exps 1–4); band power (Exps 4, 6)	TUAB	100 Hz; 1–40 Hz; first 5 min	5 s	500	50% (2.5 s stride)	preprocess_tuab.py (calls preprocess_nmt.py)
Released ViT-M (Exp 3; Exp 8 comparison)	Stroke	100 Hz; 1–40 Hz; common average reference	5 s	500	50% (2.5 s stride)	stroke_phaseB.py
FEI / GTJ pre-training	NMT (Exp 6 GTJ and fused model: + unlabelled TUAB-train)	200 Hz	5 s (FEI) / 20 s (GTJ)	1,000 / 4,000	random crops, 4 per recording per epoch	fei_pretrain.py / fei_branchC.py (Exp 6 joint: feic_joint.py); NMT input: preprocess_nmt_200hz.py
FEI, GTJ, FEI+GTJ (Exp 5)	NMT	200 Hz; z-scored only, no band-pass	20 s	4,000	none	feic_nmt_anchor.py
Band power (Exp 5)	NMT	200 Hz; z-scored only; power in 0.5–45 Hz bands	whole recording (Welch)	500 per Welch segment	50% between Welch segments	baseline_bandpower.py, act0_nmt_bandpower.py
FEI, GTJ, FEI+GTJ (Exp 6)	TUAB	200 Hz; 0.5–40 Hz; first 20 min	20 s	4,000	none	tuab_feic.py
FEI, GTJ, FEI+GTJ (Exp 8)	Stroke	200 Hz; common average reference, no band-pass; spectrum matched to NMT	20 s	4,000	none	stroke_phaseB_clean.py
BSI and spectral features (Exp 7)	Stroke	1,000 Hz native; A2 reference; 50 Hz notch; 0.5–100 Hz	2 s epochs	2,000	none (epochs over ± 100 μ V dropped)	stroke_phaseB.py

3.4 Core Source Code and Reproducibility

All the pre-training, evaluation, statistics and plotting scripts used for all the numbers and figures in this report are organized in a single script per experiment in the repository github.com/mashfiqurrahman2020-coder/eeg-jepa-act-0. Table 5 indicates which script was used to generate each report section. The one check that we used most is shown below: compare the score of a pre-trained model to a run of the same code path using a random initialisation control.

Table 5: Map from report sections to the script that generated the numbers in the report.

Script	Purpose
preprocess_nmt.py	The paper's §4.1 pipeline (crop, 1–40 Hz filter, resample to 100 Hz, z-score, window) for released-checkpoint input
preprocess_tuab.py	Runs TUAB through preprocess_nmt.py's §4.1 pipeline (Experiments 1–4, and the band-power baseline of Experiments 4 and 6)
preprocess_nmt_200hz.py	The 200 Hz NMT input FEI and GTJ read (recipe reconstructed after the original script was lost; reproduces all 2,417 stored files byte for byte)

Script	Purpose
fei_pretrain.py	FEI model, frequency masking, pre-training loop, 5-fold CV driver
fei_branchC.py	GTJ model, spectrogram construction, past-future JEPA, fusion CV driver
feic_joint.py	FEI and GTJ re-pre-trained on NMT plus unlabelled TUAB-train (Experiment 6's GTJ and fused rows)
feic_nmt_anchor.py	FEI, GTJ and FEI+GTJ on NMT, seed-0 5-fold CV (Experiment 5)
act0_nmt_rand_controls.py	Random-init twins of FEI, GTJ and FEI+GTJ on Experiment 5's folds (5 seeds)
act0_nmt_pretrain_replicate.py	Two further pre-training runs of FEI and GTJ per fold, with new seeds (Experiment 5)
act0_nmt_stats.py	Nadeau–Bengio corrected t-tests with Holm adjustment for Experiment 5 (Table 20)
baseline_bandpower.py	NMT relative band power at 200 Hz (Experiment 5)
act0_nmt_bandpower.py	NMT band power on Experiment 5's exact folds, with paired tests
tuab_bandpower.py	Hand-crafted relative band-power baseline on TUAB's 100 Hz frames (Experiments 4, 6)
tuab_released_eval.py / tuab_cached_eval.py	Released ViT-M/ViT-B frozen evaluation on TUAB (Experiment 1)
tuab_finetune_fast.py	ViT-M fine-tuned from the released weights and trained from scratch, with the authors' recipe (Experiment 2)
tuab_vitm_linear.py	Linear probe on frozen released and random-init ViT-M (Experiment 4; the ViT-M row of Experiment 6)
tuab_collapse_check.py	Token-std / cross-recording-cosine collapse diagnostics (Experiment 3)
tuab_feic.py	FEI+GTJ transfer evaluation on TUAB, with random-init controls (Experiment 6)
tuab_fei_reps.py	FEI's NMT-only TUAB gain over five pre-training runs, two of them pre-trained here, with the multi-run bootstrap (Experiment 6)
tuab_feic_joint_reps.py	Two seeded re-runs of Experiment 6's adapted (NMT+TUAB) FEI and GTJ pre-training, scored as in tuab_feic.py
stroke_phaseB.py / stroke_phaseB_cv.py	BSI baseline + LOSO / 5-fold×100 / leave-pair-out CV machinery
stroke_phaseB_clean.py	Spectrum-matching fix and its input/embedding health gates (Experiment 8)
stroke_phaseB_foldlocal.py	Stroke spectral-matching filter refitted inside each LOSO fold (Table 19)
stroke_phaseB_reps.py	Experiment 8 scored with Experiment 5's two seeded pre-training re-runs
act0_repro_manifest.py	Fold, subset and cohort lists; checkpoint hashes; environment (Appendix §11)
act0_raw_hashes.py	SHA-256 of every raw recording (Table 21)
act0_metrics.py	Shared metric definitions and bootstrap confidence intervals
act0_full_metrics.py	BAcc, AUC-PR, F1, sensitivity and specificity columns, recomputed from each experiment's saved predictions
act0_make_results.py	Experiments 1, 2 and 4's Mann–Whitney, paired t- and Wilcoxon tests, from the saved results

Script	Purpose
failure_axes.py	Per-recording failure-axis analysis from saved out-of-fold predictions

One formula named in Table 5 but never actually shown: the classical band-power baseline defining each “band power” row in Experiments 4 and 6. The power spectral density is estimated for each channel with Welch's method (already filtered, already windowed signal), summed in the five fixed bands and divided by the sum of all bands — this gives a relative power (a channel's spectral shape after removing the absolute amplitude, making the comparison fair on top of the already per-recording z-scored input signal, §3.3). The NMT band-power arm for Experiment 5 applies the same construction to NMT's own 200 Hz input, performing Welch across the entire recording, instead of per frame. Bands: 0.5-4, 4-8, 8-13, 13-30, 30-45 Hz, normalised by total 0.5-45 Hz power.

```
BANDS = [(1, 4), (4, 8), (8, 13), (13, 30), (30, 40)] # delta/theta/alpha/beta/gamma; data
already 1-40 Hz band-passed
FS = 100 # Hz, matching the shared
preprocessing pipeline (Code Snippet 3)

def rbp(path):
    x = np.load(path) # (119, 19, 500): 5 s frames, 50%
    overlap
    f, p = welch(x, fs=FS, nperseg=250, axis=-1) # Welch PSD per frame
    p = p.mean(0) # average over the 119 frames ->
    (19, freqs)
    bp = np.stack([p[:, (f >= lo) & (f < hi)].sum(-1) for lo, hi in BANDS], -1)
    return (bp / bp.sum(-1, keepdims=True)).ravel() # (95,) relative band power = 19
    channels x 5 bands
```

Code Snippet 4: Feature to use for every TUAB band-power number, used in the classical band-power baseline, in Experiments 4 and 6 (code/tuab_bandpower.py, simplified).

```
def fit_probe(Xtr, ytr, Xte, yte):
    """The ONE probe used for every encoder in this report: StandardScaler + balanced
    LogReg, C=1."""
    sc = StandardScaler().fit(Xtr)
    clf = LogisticRegression(max_iter=2000, class_weight="balanced",
    C=1.0).fit(sc.transform(Xtr), ytr)
    s = clf.decision_function(sc.transform(Xte))
    return roc_auc_score(yte, s), balanced_accuracy_score(yte, (s > 0).astype(int))

# The habit that runs through every experiment in this report:
# score_pretrained = fit_probe(embed(pretrained_encoder, ...))
# score_random = fit_probe(embed(RANDOM-INIT, same architecture, same pipeline))
# -> only trust "pre-training helped" if score_pretrained beats score_random, with a
stated p-value.
```

Code Snippet 5: The single shared evaluation probe, and the random-init-control discipline applied to each pre-trained encoder in this report (the one exception is stated below in Experiment 1).

4 Implementation

4.1 Description

Nothing was combined before it was tested on its own. The FEI was built and unit-tested separately during a few weeks prior to the mid-term break (around week 7): its self-test shows that masking a signal's 10 Hz component indeed nulls a 10 Hz test tone, and that the JEPA loss decreases over a few optimisation steps when using a synthetic batch, before any real EEG data gets to it. GTJ got the same treatment in week 8. First, it has a self-test to check the shapes of spectrograms; second, it has a test for loss drops; third, it has a test to check that the past half of a spectrogram is not numerically identical to the future half; otherwise “predict the future” would be trivially solvable and would have no meaning.

Before either of the encoders started running, one of the data-loading problems appeared – one of the earliest versions of the NMT loader loaded all recordings into memory, which caused a 24 GB out-of-memory error. Switching to a memory-mapped loader fixed that (down to about 2 GB) and, as a side effect, training was

about ten times faster, because the GPU no longer had to wait for disk reads. It should have been done from the get-go.

After completing the self-tests and single-branch NMT cross-validation (Experiment 5) for both FEI and GTJ, both were fused together to form FEI+GTJ. Each subsequent addition was built on its own first, followed by integration. The band-power and BSI baselines were verified with well-known clinical effects prior to using them as a comparative reference (in Experiment 7, the pipeline-soundness gate).

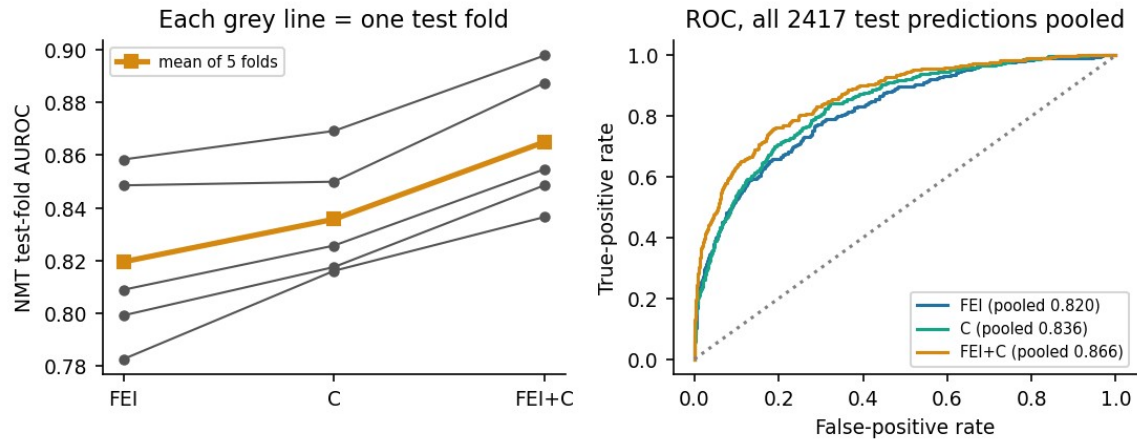


Figure 12: Fusion of FEI and GTJ into FEI+GTJ on NMT — each grey line is one CV fold; every fold's AUROC is higher when the two branches of the model are fused together as compared to the single branch.

5 Design Analysis and Evaluation

5.1 Novelty

FEI and GTJ are not in and of themselves new architectural concepts. Removing random frequency components of a time series is already a part of the augmentation pre-training for time-frequency-consistency contrastive learning [14]; FEI's twist is to predict (in embedding space) how the momentum target encoder sees the masked copy. The prediction of future signal from previous signal is already achieved by contrastive and autoregressive predictive coding for speech [15, 16] and contrastive predictive coding has already been applied to clinical EEG such as for pathology detection [17]. Two recent EEG papers are even closer: one pre-trains by autoregressively predicting future spectro-temporal content [18], and a 2026 pre-print that appeared during this project predicts future latent EEG states using an EMA target encoder [19].

We discovered that a particular publicly released EEG foundation-model checkpoint is representation collapsed. We could not find this diagnosis published at the time. Token standard deviation and cross-recording cosine similarity make it easy to repeat the check on that checkpoint in minutes.

Our working 2-branch replacement is FEI+GTJ. An earlier design of the GTJ was to mask out regions of the spectrogram and have the encoder fill them in, and Ahmed discovered that it could be solved almost entirely from each window's average spectral content: it was learning very little more than the band-power baseline already described in this report. The design was changed to a past-predicts-future one, in which the future half is never even presented to the context encoder, and that closed the shortcut. Three datasets and two clinically different tasks are used to validate the resulting pair end-to-end with paired statistics.

However, GTJ later failed on another dataset due to NMT's input statistics being carried over by the BatchNorm layers in the model. We identified that failure, and fixed it without labels. The fix is only applied to the cohort it was trained on but it could extend to other EEG self-supervised encoders that normalise like this.

5.2 Design Considerations

5.2.1 Considerations to public health and safety

A model that says that an EEG recording is “abnormal” has a direct impact on patient safety. The false negative result may lead to a delayed diagnosis. A false positive may lead to additional tests being performed, or to undue concern. There is a non-uniform distribution of error rates across age groups, as shown in our failure-axis analysis (§5.3.4) – FEI flags healthy young children as abnormal at more than six times the rate that it flags healthy adults. No system should have that large of a gap, regardless of what its AUROC is, near a real clinical decision for a paediatric patient.

5.2.2 Considerations to environment

All models are deliberately tiny (1.15M parameters for FEI, 1.88M for GTJ, including both the online and the EMA target copy; Tables 1–2), and trained for tens (not hundreds) of epochs on a single consumer GPU, instead of on the institutional clusters used for pre-training CBraMod and LaBraM. Partly this was just what hardware we had. However, it's also a true result: a model sized to match its dataset, instead of scaled up regardless of how much data is available, already outperforms a classical baseline, and with the unlabelled recordings of TUAB added to pre-training it scores close to the released large-scale checkpoint under the same linear probe on TUAB (0.862 vs 0.838, not significantly different).

5.2.3 Considerations to cultural and societal needs

Most of this work is based on NMT, which was collected specifically in Pakistan to provide a presence of under-represented, lower resource clinical populations in EEG machine-learning research. Most large datasets, including TUAB, are from hospitals that are well-resourced, typically in North America and Europe. Demonstrating the capability of a small encoder trained on this dataset to transfer its pre-training benefit to another hospital's data (§5.3.3, Experiment 6) is one small piece of evidence that self-supervised EEG models are not restricted to high-resource-setting data. The third task here is stroke, one of the most important causes of long-term disability in the world, and low-cost, automatable screening is most important where specialist neurological review is most difficult.

5.3 Investigations

5.3.1 Design of Experiment

All experiments in this report are conducted in the same way. Any claim that pre-training “helped” has to be paired with the same architecture with random untrained weights and executed by the same evaluation code. We provide a confidence interval or associated p-value with every formal inference of comparison between two models: bootstrap resampling for independent recordings; paired tests for same-subset comparisons (for cross-validation folds, the training sets overlap, so we use the Nadeau–Bengio corrected resampled t-test [20] with Holm's adjustment [21] for a family of comparisons); and label-permutation tests for small cohorts. Experiments are run in order of increasing domain shift: validate on the data on which the encoder was trained first (NMT), then transfer it, frozen, to a different hospital's data (TUAB) and only then use it on a structurally different clinical task that the encoder was never designed for (stroke vs. healthy). For the TUAB step, all three encoders are pre-trained using NMT only, and then pre-trained with the addition of TUAB's unlabelled recordings for GTJ and the fused model (Experiment 6). It's much weaker to say a model passes the 1st and easiest step than to say it passes all three.

This design has two recurring checks that are, in effect, this report's ablation study: the random-init control run alongside almost every pre-trained encoder (removing pre-training as the only variable, with architecture and evaluation code held fixed), and Experiment 4, comparing the results across feature/probe choices (removing or swapping the encoder itself, with the classifier held fixed). This project does not have a single dedicated ablation table, but the ablations are spread throughout the experiments below in a location where each experiment is most informative.

5.3.2 Data Collection

The size, role and any licensing caveats for all three datasets are described in Table 3, in §3.3 above, which outlines the process of data set acquisition and preparation.

5.3.3 Results and Analysis

AUROC is used in all of our work. It doesn't care about the decision threshold, as that is relevant because we never did decide on one. It is accompanied, if applicable, by other metrics such as balanced accuracy and AUC-PR. Unless otherwise indicated, “±” represents the standard deviation of the cross-validation folds or training subsets, and brackets represent a 95% bootstrap confidence interval.

Formally, writing TP/FP/TN/FN for true/false positives/negatives at a given threshold, sensitivity (true-positive rate) is $TPR = TP / (TP + FN)$ and specificity's complement (false-positive rate) is $FPR = FP / (FP + TN)$; **AUROC** is the area under the curve traced by (FPR, TPR) as the decision threshold sweeps over every possible value, equivalently the probability that a randomly chosen abnormal recording is scored higher than a randomly chosen normal one. **Balanced accuracy** fixes a single threshold (predicted probability $p > 0.5$ throughout this report, equivalently decision_function $s > 0$ for the logistic-regression probe of Code Snippet 5) and averages the two classes' recall so neither is drowned out by the other's size: $BAcc = \frac{1}{2} \cdot (TPR + TNR)$, where $TNR = TN / (TN + FP)$. **AUC-PR** is the same sweep-every-threshold idea as AUROC but traces precision = $TP / (TP + FP)$ against recall = TPR instead of FPR against TPR, which makes it more sensitive to performance on the minority (here, abnormal or stroke) class than AUROC is.

Experiment 1 — Frozen reproduction of the paper's TUAB result.

The simplest question first: load the authors' own released checkpoints (SHA-256 hashes in Table 18, verified against the Hugging Face file records) into the authors' own evaluation heads, run them on TUAB v3.0.1's 276 recording evaluation split that the paper reports on (files from our mirror copy, §5.5.3), and see if we get anywhere near the authors' reported AUROC of 0.877 (ViT-M) / 0.879 (ViT-B). To rule out our code as the issue, Maruf diffed our pipeline against the authors' public repository (github.com/amir-hojjati/eeg-vjepa) at commit c739fad5ad57ec4439a094343901a245edf611ec: 43 of 48 compared files were byte-identical. The other 5 contain straightforward and well commented additions of ours to the original code: a thermal guard, a fixed evaluation seed, an optional alternate head, an optional learning-rate scale, balanced accuracy among the metrics — no change to the computation at their default values.

The same evaluation heads are used throughout the experiments below: a Linear Probe, the paper's own §4.2 Attention Classifier, and its Fig-3 Attentive Probe [1], each of which takes the output of a frozen (or, in two cases, fine-tuned) encoder and converts it to a normal/abnormal decision.

Downstream Linear Probe Architecture

Used for frozen evaluation on TUAB

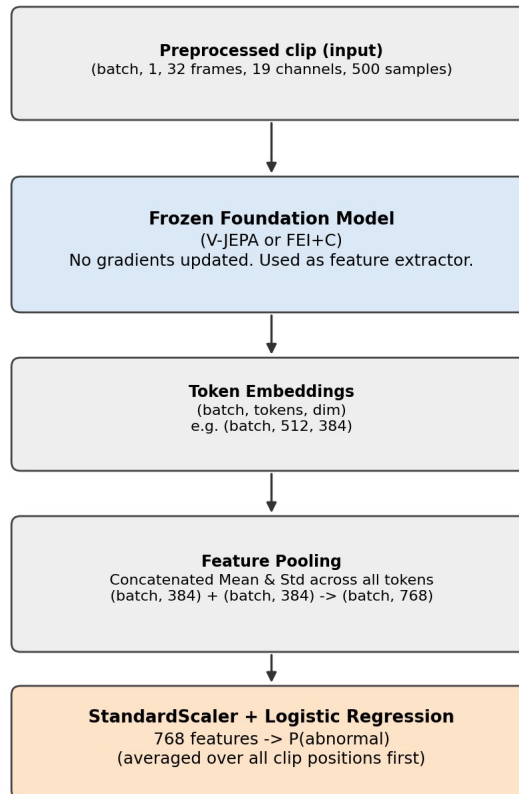


Figure 13: Linear Probe — mean and std dev of all tokens (or raw embedding for FEI/GTJ) then a StandardScaler + logistic regression.

The simplest of the three is Linear Probe which takes mean and STD of 512 output tokens (384+384=768 numbers) averaged over each clip position in a recording for ViT-M; for FEI/GTJ it is just their 256(+256)-number embedding. In either case, a single `StandardScaler` + class-balanced `LogisticRegression` is placed on top (the same lightweight head that is used for each of the encoders in this report); this means that no model is helped by a stronger classifier than the others.

Downstream Attention Classifier (Fine-Tuning)

Used for end-to-end fine-tuning on TUAB

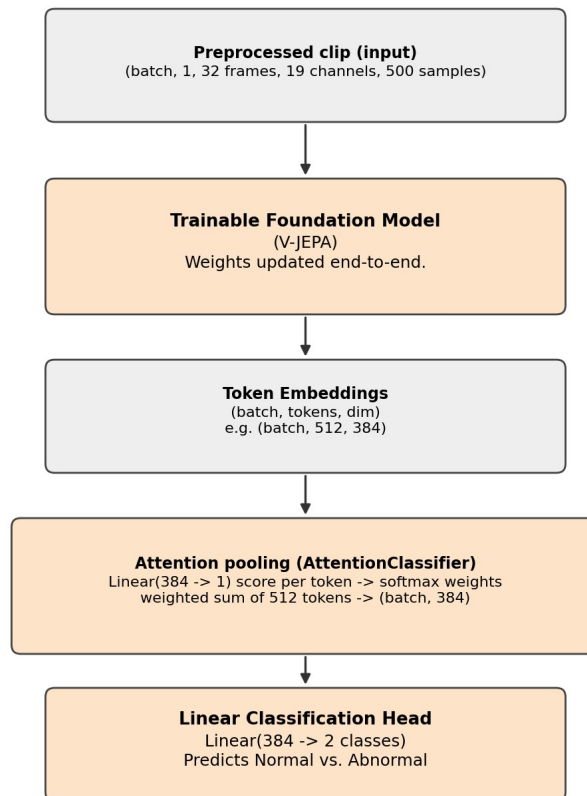


Figure 14: The §4.2 Attention Classifier in the paper is a learned score per token, a softmax normalisation to weights, a weighted average, and then a linear classifier [1].

The §4.2 Attention Classifier (`AttentionClassifier` in the authors' own code) is the head the paper itself evaluates with, and it is the only one of the three heads that was used for fine-tuning. It's just two learned layers: A linear(384 → 1) layer scores each of the 512 tokens, softmax turns those 512 scores into weights that sum to 1, the tokens are averaged with those weights to get a single 384-number vector, so the head learns which tokens to listen to, and a final Linear(384 → 2) reads off the normal/abnormal decision. It is used as the head trained end-to-end when we fine-tune the network or train it from scratch (Experiment 2) [1], and also when it is frozen (on the released checkpoint).

Attentive Probe (AttentiveClassifier, Fig-3 head)

Frozen evaluation on TUAB only; ViT-M sizes shown

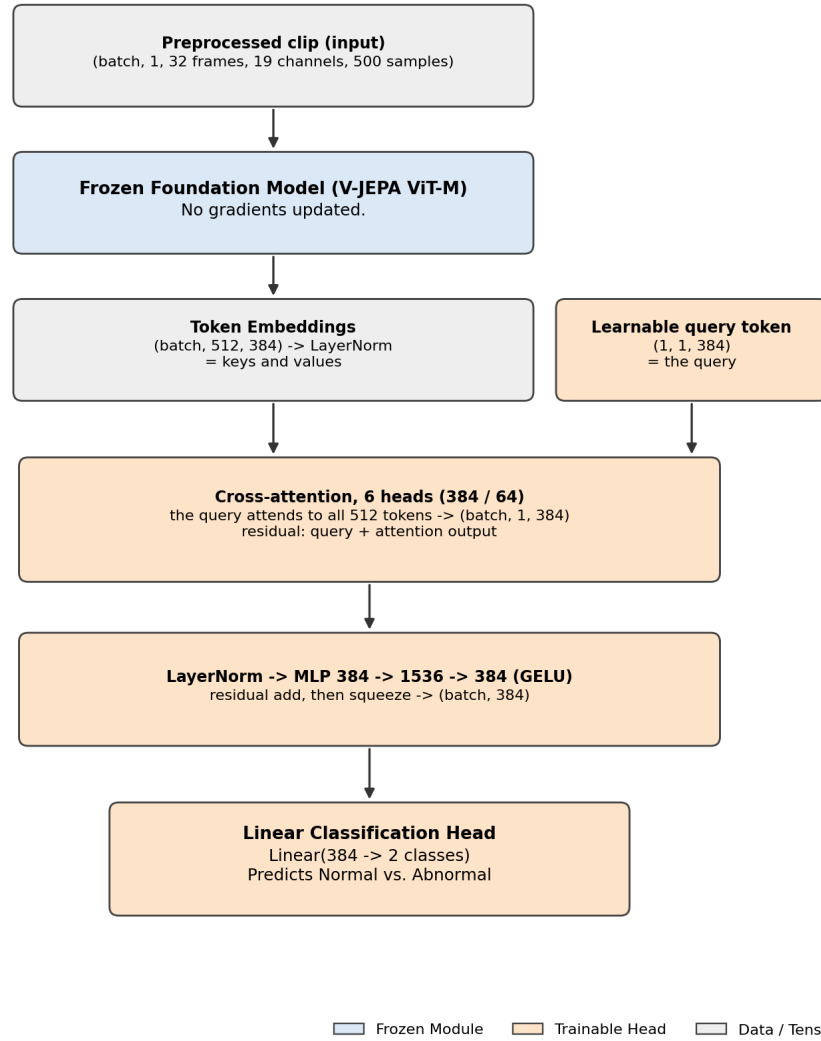


Figure 15: In the paper, the Fig-3 Attentive Probe is a learnable query token that attends to all 512 tokens using full multi-head cross-attention, followed by an MLP [1].

The larger, frozen-only one is the Fig-3 Attentive Probe (‘AttentiveClassifier’) with one learnable query token attending to all 512 tokens via a full cross-attention block (6 heads of 384/64 for ViT-M) that feeds an MLP before the final classification layer. This has significantly more trainable weights than the §4.2 head, which is why it’s the more forgiving of the two frozen heads for the released checkpoint, and also why in Experiment 4 below we try our own, even simpler linear probe on the same frozen features, to make sure neither of the authors’ heads is the limiting factor [1].

Table 6: The three evaluation heads throughout this report, including number of trainable-parameters. The Linear Probe is a scikit-learn LogisticRegression, not a PyTorch neural network, so thus the “parameters” are fitted coefficients, not gradient trained weights, while the other two are PyTorch modules from the authors’ own upstream/eeg-vjepa code [1].

Head	Sits on	Trainable parameters	Used with
Linear Probe	ViT-M mean+std (768-dim)	769 (sklearn coefficients + intercept)	Released/random ViT-M (Experiments 1, 4, 6)
Linear Probe	FEI alone (256-dim)	257	FEI pre-training ablations
Linear Probe	FEI+GTJ concat (512-dim)	513	Combined FEI+GTJ runs (Experiment 6)

Head	Sits on	Trainable parameters	Used with
§4.2 AttentionClassifier	ViT-M tokens (384-dim)	1,155	Released/random ViT-M, frozen and fine-tuned (Experiments 1, 2)
Fig-3 AttentiveClassifier	ViT-M tokens (384-dim)	1,775,618 (1.78M)	Released/random ViT-M, frozen only (Experiment 1)

Table 7: Experiment 1 — released-checkpoint TUAB AUROC/BACC/AUC-PR (final epoch; BACC and AUC-PR at the default 0.5 threshold, the same convention used everywhere in this report) vs random-init controls and the paper's reported AUROC (*best-epoch is optimistic, chosen by looking at the test set, and shown only for reference; ViT-B was not run with a random-init twin) [1]. Runs: one evaluation head trained per TUAB training subset (s0–s4); each random-init run (s0–s2) also has its own random weights, seeded with its subset number.

Encoder + head	Runs	AUROC	BACC	AUC-PR	Best-ep. AUROC*	Paper's AUROC
ViT-M, §4.2 head (paper's own)	5	0.671 ± 0.020	0.626 ± 0.023	0.685 ± 0.021	0.737	0.877
ViT-M, Fig-3 head	5	0.713 ± 0.025	0.659 ± 0.033	0.717 ± 0.035	0.784	0.877
Random-init ViT-M, §4.2 head	3	0.787 ± 0.006	0.704 ± 0.004	0.792 ± 0.010	0.844	—
Random-init ViT-M, Fig-3 head	3	0.760 ± 0.016	0.707 ± 0.022	0.756 ± 0.006	0.839	—
ViT-B, §4.2 head	5	0.662 ± 0.030	0.598 ± 0.020	0.654 ± 0.024	0.715	0.879
ViT-B, Fig-3 head	5	0.762 ± 0.031	0.687 ± 0.029	0.753 ± 0.032	0.814	0.879

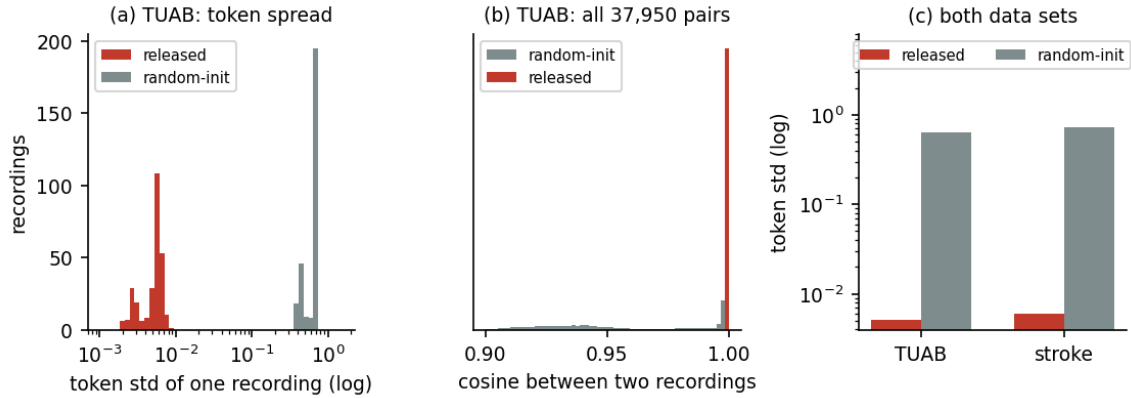


Figure 16: Direct evidence of representation collapse is that the per-recording output spread (token std) of the released checkpoint was about 125× smaller than a random-weight twin's on both TUAB and stroke (never seen in pre-training), and the cosine similarity between recordings, regardless of diagnosis, was nearly perfect (mean 0.999982). Token spread is the criterion: random weights also achieve a cosine of 0.9991 on the stroke cohort.

All released-checkpoint configurations are about 0.1-0.2 AUROC behind the paper. More impressively, a random, untrained ViT-M outpaces the released, pre-trained ViT-M: by 0.116 AUROC on the §4.2 head, 0.046 on the Fig-3 head (one-sided Mann–Whitney over 5 released and 3 random-weight runs: $p = 0.018$, the smallest these counts can allow, and 0.036; two-sided, 0.036 and 0.071, so the Fig-3 gap alone would not pass a two-sided test). This gap is observed from the very first epochs in training curves of the authors' own recipe, over all 500 epochs. More training was not the solution. Experiment 3 demonstrates why: the output of the released encoder changes relatively little from one recording to the next, meaning that a top layer classifier has

little to go on.

Experiment 2 — Fine-tuning vs. training from scratch.

If the released weights are collapsed, does starting from them still help once the whole network is fine-tuned end-to-end, rather than left frozen? The paper's AUROC is 0.885 after full fine-tuning. We fine-tuned the whole ViT-M plus the §4.2 head twice (once from the released weights, once from random weights) using the same 5 training subsets, with the authors' own recipe (500 epochs, batch size 2, learning rate 1e-3, cosine decay). A “training subset” here, and in every TUAB experiment below, is one of five distinct samples drawn from TUAB's 2,717-recording training pool (2,993 total minus the 276-recording Experiment 1 eval set) – mirroring the paper's own five-subset protocol, the exact recording membership of which the authors never published (§5.4).

Table 8: Experiment 2 — final-epoch AUROC/BAcc/AUC-PR (mean $\pm$ sd over 5 subsets of training data), fine-tuning from released checkpoint vs. fine-tuning from random weights, on the same 5 subsets of training data (released fine-tuning recipe in the paper).

Start point	AUROC	BAcc	AUC-PR
Fine-tune (released weights)	0.755 $\pm$ 0.053	0.684 $\pm$ 0.043	0.763 $\pm$ 0.055
From scratch (random weights)	0.768 $\pm$ 0.114	0.699 $\pm$ 0.077	0.795 $\pm$ 0.099

The two starting points are not significantly different (paired t-test and Wilcoxon both $p = 0.81$); any two of the five subsets share 98-134 of their 546 recordings, which both tests ignore, so allowing for that overlap would make the uncertainty wider, not narrower. From-scratch training even wins on 3 of 5 subsets. There's no measurable advantage in using the released, “pre-trained” weights as opposed to starting from scratch, which is exactly what we'd expect if those weights contain no information of any value.

Experiment 3 — Confirming representation collapse.

Experiments 1 and 2 are both going in the same direction, so we tested it directly. For each recording, Maruf computed the token standard deviation (how much the encoder's output varies within one recording) and the mean cosine similarity between any two recordings' embeddings (where the value of 1.0 indicates that the encoder produces the same output for all recordings). Operationally, each recording is represented as one deterministic clip (32 frames spread over its first 96 frames), which the encoder encodes into 512 tokens of 384 dimensions; token std is the standard deviation over those 512 tokens (per dimension, then averaged across the 384), and cosine is computed between the mean-pooled 384-dimensional embeddings of two recordings, averaged across all pairs of recordings.

Table 9: Experiment 3 — collapse diagnostics on two entirely different datasets, one of these (stroke) was never seen in any format by the released model during the pre-training.

Dataset	Encoder	Token std (mean)	Mean cosine between recordings
TUAB eval (276 recs)	Released ViT-M	0.0051	0.999982
TUAB eval (276 recs)	Random-init ViT-M (3 seeds)	0.634	0.974
Stroke (15 subjects)	Released ViT-M	0.0060	0.999994
Stroke (15 subjects)	Random-init ViT-M (3 seeds)	0.717	0.9991

There's no overlap between the two groups. The largest token std measured for the released checkpoint, on any recording, was 0.0085. The smallest of the random-weight twins was measured as 0.379. So that eliminates measurement noise. The released EEG-VJEPA checkpoints give almost the same output for every input, on two entirely different patient populations. This is not altered by any preprocessing change we tested, and it holds on both datasets, so it seems to be an attribute of the released weights, not of our pipeline or of the

dataset we test on. We even checked whether our own z-scoring convention could be doing this: per-recording and per-window normalisation resulted in the same collapsed number (token std 0.0059, cosine 1.0000) on the same 120 recordings. We fixed no numeric cut-off for collapse in advance – the criterion is separation from every random-weight twin, and a gap of 0.0085 against 0.379 doesn't require a cut-off to read. The deciding measure is the token spread. Cosine alone is not: on the 15-subject stroke cohort, even the random weights result in a cross-recording cosine of 0.9991 (Table 9).

So, there's a clear question here: how is collapse even possible when there's an anti-collapse mechanism in place? Their public implementation does include a variance-hinge term, ``relu(1 - std)``, which penalises the predictor's output for not having enough variance between the different patches of a clip it is being asked to predict, which is the same idea that underpinned FEI's own BYOL-like design. Maruf scoured through their published training configs and located a possible culprit: The weight of that term, called ``reg_coeff``, is set to 0.0 in the config file that produces the released ViT-M checkpoint (``configs/pretrain/cluster_vitl16_EEG.yaml``). Switched off. What remains is just a plain unnormalised L1 loss between the output of the predictor and a layer normalised target; a near-constant output can satisfy an L1 loss almost for no cost. We have not re-run their pre-training with this term switched back on, so this is the most probable explanation we found in their own code and config – not an independently re-verified causal test. What is shown is the collapse, itself (Table 9) and the setting in the released config; that the setting caused the collapse is a hypothesis. Another, untested, candidate is also in the same code: pre-training saves the checkpoint with the lowest training loss as ``jepa-best``, and the released file is that checkpoint (epoch 353). If the loss is one that a near-constant output can satisfy, the lowest loss checkpoint could very well be the most collapsed.

Experiment 4 — The simplest possible probe, plus a classical baseline.

Perhaps there is some helpful information in the released features that the paper's classifier simply isn't learning from. We investigated the simplest alternative we could come up with: mean and standard deviation of the encoder's output of tokens (768 numbers total) which we fed to StandardScaler and logistic regression, and, for comparison, a purely classical, hand-crafted feature set consisting of relative band-power, with no learned encoder whatsoever.

Table 10: Experiment 4 — AUROC (a simple linear probe recovers more signal than the paper's own head, but still falls short of the paper's 0.877 and is not clearly better than random weights) plus the full operating point at the full-training-set size (BAcc/AUC-PR/Sens/Spec at the default 0.5 threshold; random-init ViT-M has no single full-train run, hence the dashes).

Features	AUROC, subsets s0–s4	AUROC, full-train	BAcc, full-train	AUC-PR, full-train	Sens, full-train	Spec, full-train
Band power (hand-crafted, no encoder)	0.730 ± 0.014	0.757	0.670	0.718	0.754	0.587
Random-init ViT-M	0.784 ± 0.023	—	—	—	—	—
Released ViT-M	0.805 ± 0.013	0.838 [0.787, 0.884]	0.762	0.846	0.730	0.793

This simpler probe recovers more signal than the paper's own heads (0.805 ± 0.013 as compared to 0.671–0.713 for the paper's heads on the same five training subsets; 0.838 on the full training set) and both variants of ViT-M outperform the classical band-power baseline. But the released checkpoint is still not significantly different from a random-weight twin (one-sided Mann–Whitney $p = 0.196$, 5 released vs. 3 random-weight runs) and its full-training-set score of 0.838 is still 0.039 points behind the paper's 0.877. Of the three frozen heads we ran, this linear probe gives the released checkpoint its highest TUAB eval AUROC; we picked it after seeing all three results, which can only flatter the released model, and it is the bar Experiment 6 compares FEI+GTJ against.

Read together with Experiment 1's per-head comparison (Table 7), this is the ablation sweep over all that lies in between a frozen embedding and final decision: encoder (band power / random-init ViT-M / released ViT-M) crossed with head (Linear Probe / §4.2 AttentionClassifier / Fig-3 AttentiveClassifier). Swapping either one does move the number, but neither swap closes the gap with the reported 0.877 from the paper; the

shortfall behaves as a property of the released weights, not of which head reads them.

One caveat on the band-power baseline applies through Experiment 6 – the 0.730–0.757 numbers above use relative spectral power only, without consideration of the absolute amplitude of the signal in each channel. We did a separate sanity check, primarily to verify that the relative-power implementation was not in itself buggy, but not to move towards the strongest possible classical feature, and found that adding absolute (log) power on a longer window of 20 minutes (not this report's 5 minutes) raised the baseline to 0.808, which narrowed, without closing, the gap to FEI+GTJ and the released ViT-M's linear-probe score. The weaker, 5 minute relative-power version is kept in all of the TUAB experiments so that they stay comparable; Experiment 5 instead has its own NMT band-power arm, scored on the same folds as the encoders it is compared with. Read “beats band power” not as beating the most powerful classical feature set we found, but as beating this specific baseline that is meant to be deliberately simple.

Experiment 5 — FEI+GTJ validated on NMT, the data it was trained on.

Given the inability to use the released model as a useable pre-trained encoder, we tested FEI+GTJ for subject-independent, seed-0, five-fold cross-validation over all 2,417 NMT recordings. The encoders for each fold are only pre-trained with that fold's own training data, and no test recording (labelled or not) is seen during pre-training. Every recording's fold is listed in the file docs/act0_repro/nmt_folds_seed0.csv, which is the StratifiedKFold split of NMT's train and eval recordings concatenated in that order, stratified by label. NMT holds one recording per participant, so the split is also subject-independent.

Table 11: Experiment 5: FEI, GTJ and FEI+GTJ on NMT compared to the classical band-power baseline, seed-0 5-fold cross validation, mean $\pm$ sd over folds, full metric set with the default threshold of 0.5. Band power is scored on the same folds, using the same probe and on NMT's own 200 Hz input. The pre-trained rows are the original pre-training run; each rand- row is its random-init twin, averaged over 5 random seeds per fold.

Features	AUROC mean $\pm$ sd (5- fold CV)	BAcc	AUC-PR	F1	Sens	Spec
Band power (classical, NMT)	0.786 $\pm$ 0.009	0.720 $\pm$ 0.013	0.429 $\pm$ 0.033	0.458 $\pm$ 0.013	0.773 $\pm$ 0.025	0.667 $\pm$ 0.010
FEI alone	0.819 $\pm$ 0.029	0.729 $\pm$ 0.025	0.563 $\pm$ 0.048	0.486 $\pm$ 0.028	0.708 $\pm$ 0.049	0.750 $\pm$ 0.019
rand-FEI (random-init control)	0.810 $\pm$ 0.025	0.728 $\pm$ 0.019	0.548 $\pm$ 0.054	0.494 $\pm$ 0.025	0.675 $\pm$ 0.032	0.780 $\pm$ 0.018
GTJ alone	0.836 $\pm$ 0.021	0.752 $\pm$ 0.033	0.572 $\pm$ 0.041	0.522 $\pm$ 0.036	0.718 $\pm$ 0.064	0.786 $\pm$ 0.017
rand-GTJ (random-init control)	0.794 $\pm$ 0.009	0.717 $\pm$ 0.014	0.483 $\pm$ 0.018	0.481 $\pm$ 0.015	0.658 $\pm$ 0.032	0.777 $\pm$ 0.010
FEI+GTJ (fused)	0.865 $\pm$ 0.024	0.778 $\pm$ 0.030	0.657 $\pm$ 0.062	0.570 $\pm$ 0.034	0.725 $\pm$ 0.060	0.831 $\pm$ 0.011
rand- FEI+GTJ (random-init control)	0.834 $\pm$ 0.007	0.749 $\pm$ 0.010	0.586 $\pm$ 0.033	0.542 $\pm$ 0.014	0.658 $\pm$ 0.020	0.840 $\pm$ 0.012

When compared to FEI alone and GTJ alone in each of the five folds, fusion beats them both, which is a good indication that the two branches are learning different aspects of the signal. There are eleven comparisons shown in Table 20 (Appendix §11.3), and the five training sets overlap (any two share three-fourths of their recordings), so an ordinary paired t-test over folds would understate the variance; therefore, each p value in this experiment is the Nadeau–Bengio corrected resampled t-test [20], Holm-adjusted [21] over those eleven comparisons. The pre-trained arms are averaged over three independent pre-training runs, the original plus two additional runs, each with different seeds, whose fused means were 0.865, 0.865 and 0.852; the runs are averaged within each fold before being tested, thus the variation between different runs are folded into the fold means rather than modelled separately. On those means fusion is ahead of FEI by +0.041 ($p = 0.027$) and of GTJ by +0.030 ($p = 0.012$ before Holm, 0.082 after). Against band power on the same folds, the fused model is

ahead by +0.075 AUROC (all 5 folds, $p = 0.029$) and GTJ by +0.045 (all 5 folds, $p = 0.027$); FEI's +0.034 (4 of 5 folds) is not significant ($p = 0.45$). The random-init twins alter the meaning of the margin. Most of the benefit of the fused model is due to its architecture and input representation, as an untrained fused encoder already achieves 0.834 ± 0.007 , which is 0.048 higher than band power (all 5 folds, $p = 0.026$). Pre-training adds +0.027 on top (three-run mean 0.861 against 0.834), better in all 5 folds but not significant once corrected ($p = 0.15$ before Holm, 0.45 after); the gain is +0.036 for GTJ (all 5 folds, $p = 0.019$ before Holm, 0.12 after) and only +0.010 for FEI (3 of 5 folds). This band-power arm is equivalent to the relative-band-power feature of Table 10's but recomputed using NMT's own input of 200 Hz, with its bands running from 0.5 to 45 Hz.

Experiment 6 — FEI+GTJ on a different hospital's data (TUAB): NMT-only and adapted.

FEI+GTJ has been built and validated entirely on NMT (Pakistan). In Experiment 6 it is tested on TUAB (USA): different hospital, different hardware, different patient population. Two settings are separated. NMT-only transfer: the encoder is pre-trained only on NMT, and it has not received any TUAB recording (whether labeled or unlabeled) before its frozen features are probed. This is sometimes referred to as zero-shot, but only the encoder's pre-training is: the linear classifier on top is trained on labelled TUAB recordings, so it tests the transfer of the representation across datasets, not zero-shot classification. Target-domain unlabelled adaptation: the encoder is re-pre-trained using NMT and the unlabelled training recordings from TUAB (never the eval set, never a label), and then is probed the same way. In both contexts the linear probe is trained on TUAB's labelled training subsets and tested on the 276-recording eval set, as in Experiment 4.

Both settings are included in Table 12. The clean version of the cross-dataset test for representation transfer is FEI's NMT-only row. Not surprisingly, GTJ does not fare well on the NMT-only transfer: it performs below its own random-init twin (0.840 vs 0.863 full-train AUROC), consistent with TUAB's differently filtered spectrum being out of distribution for an encoder trained on NMT's. The adapted rows therefore re-pre-train GTJ, and the FEI inside the adapted fused model, on NMT + unlabelled training recordings from TUAB. The NMT-only FEI and the FEI within the adapted fused model are not the same weights, but rather two separate pre-trained encoders. An untrained encoder is identical in both settings, so each random-init row is used as a control in both cases.

Table 12: Experiment 6: frozen FEI, GTJ and FEI+GTJ on TUAB. NMT only rows: rows that are encoders trained on NMT only; adapted rows: rows that are encoders trained on NMT plus unlabelled training recordings from TUAB. Matched random-init controls for each arm; full-train operating point (BAcc/AUC-PR/Sens/Spec using the default threshold of 0.5) and AUROC.

Features	AUROC subsets s0–s4	AUROC full-train	BAcc full-train	AUC-PR full-train	Sens full-train	Spec full-train
Band power (classical)	0.730 ± 0.014	0.757	0.670	0.718	0.754	0.587
Released ViT-M (linear probe, Table 10)	0.805 ± 0.013	0.838	0.762	0.846	0.730	0.793
FEI (NMT-only)	0.835 ± 0.015	0.872	0.785	0.871	0.730	0.840
rand-FEI (random-init control)	0.790 ± 0.020	0.813	0.743	0.829	0.698	0.787
GTJ (NMT-only)	0.805 ± 0.005	0.840	0.768	0.835	0.722	0.813
GTJ (NMT+TUAB -train, adapted)	0.811 ± 0.012	0.858	0.767	0.853	0.714	0.820
rand-GTJ (random-init control)	0.819 ± 0.014	0.863	0.779	0.848	0.738	0.820
FEI+GTJ (NMT-only)	0.821 ± 0.006	0.857	0.792	0.852	0.730	0.853
FEI+GTJ (NMT+TUAB -train, adapted)	0.821 ± 0.012	0.862	0.764	0.855	0.722	0.807

Features	AUROC subsets s0–s4	AUROC full-train	BACC full-train	AUC-PR full-train	Sens full-train	Spec full-train
rand-FEI+GTJ (random-init control)	0.834 ± 0.014	0.861	0.761	0.860	0.722	0.800

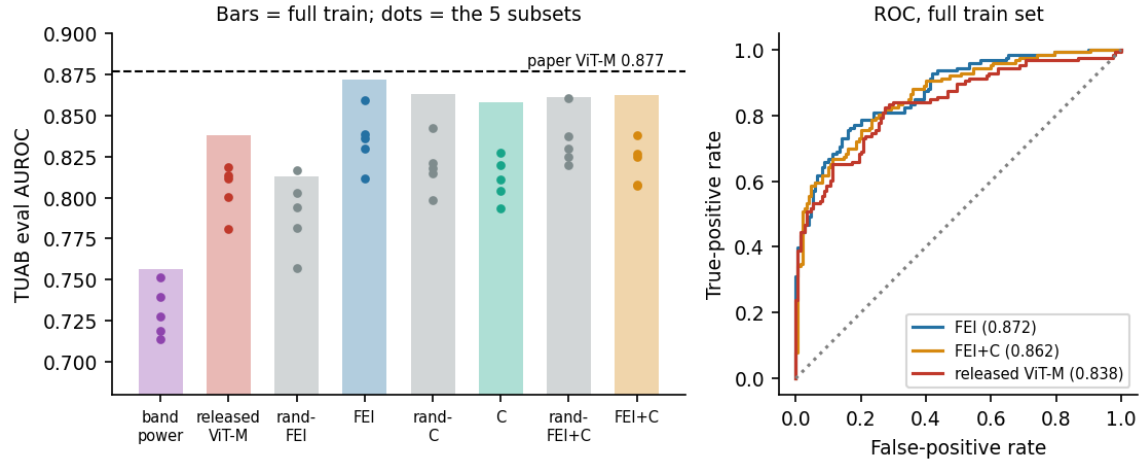


Figure 17: FEI+GTJ's TUAB transfer alongside the released ViT-M, band power and every random-init control: the five training subsets are shown as dots, and the full 2,717-recording training set as bars.

Adapted, FEI+GTJ scores above the released ViT-M and clearly beats band power (0.862 vs. 0.838 vs. 0.757 on the full training set), though the gap between FEI+GTJ and ViT-M (+0.024) is not statistically significant ($p = 0.147$). Two seeded re-runs of that adapted pre-training (with the same scoring) reach 0.851 and 0.855; when resampling the three runs and the test recordings together, the advantage of FEI+GTJ over ViT-M is only +0.018 (95% CI $[-0.028, 0.064]$, $p = 0.220$), which is not significant. NMT-only, the fused model scores 0.857 on the full training set (0.821 ± 0.006 on the five subsets), so the majority of the adapted model's score does not depend on the model seeing TUAB during pre-training.

FEI's NMT-only pre-training gain, however, is statistically significant on its own: FEI beats its random-init twin by +0.059 AUROC (95% CI $[0.019, 0.097]$, $p = 0.002$): this can be interpreted as meaning that what FEI learned on NMT alone carries information about abnormal EEG on TUAB that the architecture by itself does not supply (one dataset pair). That p-value resamples the 276 test recordings for one set of trained encoders (five, one per NMT training fold, which are then combined together), so it says nothing about how the gain varies across pre-training runs. We therefore repeated the NMT pre-training and scored each run on TUAB the same way, against the same random-init twins: Experiment 5's two seeded re-runs, plus two additional FEI-only runs with the same recipe and seeding, trained for this check alone (Experiment 5's analysis was fixed at three runs of both branches before these two were trained, and with no GTJ partner they cannot enter its fused comparison). Across the five runs, the gain over the twin is +0.048 on average (with a range of +0.036 to +0.059). When all three sources of uncertainty are included in a single interval (Table 24), a bootstrap resampling of the five pre-training runs, the twin's three random-init seeds and the 276 test recordings results in a mean gain of +0.048 (95% CI $[0.015, 0.108]$ $p = 0.0055$). All five runs are scored on one test set, so they show that the gain does not depend on one seed used in pre-training; they are not five independent replications of the transfer, and the between-run part of the interval is rough too, since there are five runs and three seeds.

The other half of the same table, GTJ is the more sobering. Adapted GTJ vs. its random twin is -0.005 ($p = 0.642$, not significant), adapted FEI+GTJ vs. its random twin is only +0.001 ($p = 0.455$), and NMT-only GTJ is below its twin outright (0.840 vs 0.863). Two seeded re-runs of the adapted pre-training tell the same story: GTJ -0.017 ($p = 0.879$) and -0.023 ($p = 0.944$), FEI+GTJ -0.010 ($p = 0.734$) and -0.006 ($p = 0.665$) against the same twins, and the bootstrap over the three runs, the twins' random-init seeds and the test recordings gives -0.015 (95% CI $[-0.040, 0.031]$, $p = 0.671$) for GTJ and -0.005 (95% CI $[-0.033, 0.043]$, $p = 0.431$) for FEI+GTJ. No pre-training improvement is observed for either adapted GTJ or adapted FEI+GTJ in the three runs, consistent across three initialisations, but so far not proof that none exists. FEI on its own, from the same

adapted pre-training runs, stays above its random-init twin in all three (AUROC 0.840–0.859): +0.046 ($p = 0.0085$), +0.027 ($p = 0.113$) and +0.041 ($p = 0.020$). The same bootstrap over runs, random-init seeds and test recordings yields +0.038 (95% CI $[-0.0003, 0.104]$, $p = 0.026$; the interval only just reaches zero), a little below the NMT-only runs' +0.048. That gain is not reflected in the fused comparison: the fused model's twin has random-init GTJ which, on its own, already reaches 0.863, above all three adapted FEI runs. The same collapse diagnostics as Experiment 3 show this is not collapse in disguise: on the TUAB test recordings, the adapted GTJ encoders of all three runs keep an embedding std of 0.33–0.39 (cosine 0.59–0.71), and every FEI encoder, NMT-only or adapted, re-runs included, 0.45–1.08 (cosine 0.64–0.84). NMT-only GTJ is the one compressed case (std 0.04–0.07, cosine 0.96–0.99, below every random-init twin's 0.11), much less so than the released checkpoint (Experiment 3) and in keeping with the spectrum shift described above. GTJ's spectrogram input and architecture are doing useful work on TUAB, but the pre-training itself is not the reason why—a distinction the failure-axis analysis below returns to.

Visual check — do t-SNE / UMAP maps show separate normal/abnormal clusters?

We originally proposed a 2-D visual inspection of the learned features, similar to the paper's own UMAP figure, which shows visibly separated clusters. The same type of map was made for the released ViT-M, its random-init twin, FEI+GTJ, and its random-init twin, on the TUAB evaluation set.

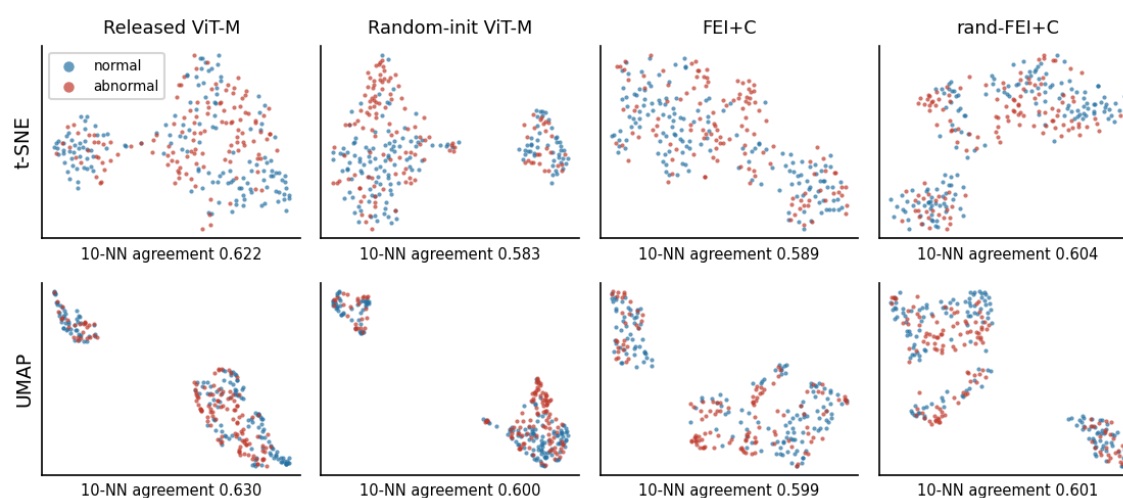


Figure 18: 2-D t-SNE map (top) and UMAP map (bottom) of 4 feature sets on TUAB, blue = normal, red = abnormal. There is no map (learned or random) that depicts the two labels in separate areas.

None of the four maps, including the two random-weight maps, have normal and abnormal recordings in distinct areas. Despite a linear probe on the same features yielding AUROC of 0.838 (released ViT-M) or 0.862 (FEI+GTJ), the 10-nearest-neighbour label-agreement statistic — proportion of 10 nearest neighbours with the same label — stays in a narrow 0.583–0.630 band for all four feature sets, on the original high-dimensional features as well as on their 2-D t-SNE/UMAP projections, only barely above the chance level of 0.502. These encoders appear to separate the two classes along a particular direction in the embedding space that a linear classifier can find, instead of by an overall 2-D layout that t-SNE or UMAP could show. Not only that, the map of the released ViT-M looks just as organised as the map of a healthy random-init one: a good reminder that a nice looking visualisation is not, on its own, evidence against collapse. That was only picked up by the direct diagnostics of Experiment 3.

Experiment 7 — Pipeline-soundness gate and the clinical BSI baseline (stroke).

Before relying on the numbers from any model on the small, 15-subject stroke cohort, Soumya first verified that the data pipeline is able to replicate a well-established classical effect — that the left-right power asymmetry (Brain Symmetry Index) of stroke patients is greater than that of healthy controls. This gate was pre-declared with a fixed significance threshold, before anyone looked at a single model's classification result; it is the gate section of runs/act0/phaseB/PREREGISTRATION.md (Appendix §11.2).

Specifically, Soumya applied the pairwise-derived BSI (pdBSI) introduced by Sheorajpanday et al. [22], a more localised modification than the single whole-head ratio of the original BSI cited above [9], which is

derived as follows: in each of the six symmetric electrode pairs (FC1/FC2, FC3/FC4, C1/C2, C3/C4, C5/C6, CP1/CP2) and for each frequency bin within the band of interest, $\text{pdBSI} = \text{mean}(|(P_R - P_L) / (P_R + P_L)|)$, where P_L and P_R are the spectral powers of the left and right hemispheres of the brain, respectively, of a specific pair; the score is the average of this ratio across all 6 pairs and all bins in the band of interest. Measuring the average of six local ratios instead of just one global ratio, makes the measure more sensitive to a focal asymmetry (expected when stroke occurs in one hemisphere), which is why it was a better option for the small cohort in this study.

Table 13: Experiment 7(a) — Pipeline-soundness gate and classical spectral features around it. The pre-declared gate was the first row (broadband BSI higher in stroke, two-sided Mann–Whitney $p < 0.05$): passed, $p = 0.003$. The pre-registration also asked for the direction of DAR, DTABR and relative delta, alpha and beta power, and all 5 of these agree; the remaining rows are reported for completeness. All features in the primary cohort move in the expected post-stroke direction, and 10 of 11 are significant there at $p < 0.05$ (the exception is BSI theta, $p = 0.113$). The sensitivity cohort, which by its own definition has a shorter window (including some task-adjacent data), confirms all BSI features but loses significance on the relative-power and ratio features; this is consistent with a noisier comparison, not with a reversed effect.

Feature	Direction in stroke	Primary median: stroke / control	Primary p	Sensitivity p
BSI 1–30 Hz (broadband)	↑	0.149 / 0.073	0.003	< 0.001
BSI delta	↑	0.109 / 0.081	0.005	< 0.001
BSI theta	↑	0.098 / 0.062	0.113	0.021
BSI alpha	↑	0.188 / 0.091	0.012	0.002
BSI beta	↑	0.142 / 0.066	0.003	< 0.001
Relative delta power	↑	0.485 / 0.214	0.008	0.068
Relative theta power	↑	0.172 / 0.106	0.036	0.016
Relative alpha power	↓	0.119 / 0.264	0.050	0.173
Relative beta power	↓	0.133 / 0.378	0.026	0.068
DAR (delta/alpha power ratio)	↑	3.407 / 0.957	0.018	0.122
DTABR ((delta+theta)/(alpha+beta))	↑	1.758 / 0.517	0.002	0.027

Table 14: Experiment 7(b) — the Brain Symmetry Index alone (no learned encoder at all) is already a strong and stable stroke-vs-control classifier under leave-one-subject-out CV.

Cohort	BSI classifier, LOSO AUROC	95% CI	Permutation p	Sens.	Spec.
Primary	0.907	[0.679, 1.000]	0.005	0.889	1.000
Sensitivity	0.938	[0.775, 1.000]	0.002	0.900	1.000

Both results serve as a sanity check to ensure the pipeline recovers the expected BSI effect, and they make BSI a tough reference baseline: alone it reaches an AUROC of 0.907, and any self-supervised encoder is tested against it, regardless of how interesting its pre-training might be.

Experiment 8 — FEI+GTJ on stroke, after fixing a distribution-shift problem.

The first attempt to implement the FEI+GTJ system on stroke data hit a snag in the form of an incorrect assumption. We had treated NMT and the stroke cohort as if they shared the same recording band (0.5–40 Hz). They don't: NMT's own EDF header has a blank prefilter field, and its real spectrum rolls off closer to 35 Hz, while the stroke side's 0.5–40 Hz filter passes content up to approximately 44–45 Hz. Soumya traced this down after GTJ's BatchNorm layers, which store NMT's input statistics, saw the stroke cohort's spectrum as wildly out of range (z-scores of 4.3 standard deviations or more) and collapsed. The same failure mode as in Experiment 3, except that this time it's caused by a failure to match the spectra, not by a training-objective problem.

The solution: a label-free, per-channel zero-phase filter that maps the stroke cohort's mean spectrum on NMT's, gated on numerical thresholds. It was not pre-registered: under the pre-registered input, FEI+GTJ had obtained 0.833 LOSO AUROC, which is worse than what its twin with random initialization obtained, 0.889; and that result is what prompted the fix. The recipe, its gates and the comparisons to run were written to runs/act0/phaseB_clean/POSTHOC.md before any LOSO run was executed on the corrected input (Appendix §11.2), and all gates passed. The largest difference between the stroke and NMT log-spectra fell from 5.81 to 0.18 decades (gate: ≤ 0.25 decades), the BatchNorm inputs of GTJ fell from $|z| \approx 4.3$ to ≈ 0.37 , and no embedding dimension remained constant across subjects. Experiment 8 is thus a post-hoc repair of a distribution shift, tested with the very same 15 subjects whose result prompted it, rather than an independent validation. Refitting the filter inside each fold (Table 19) rules out test-subject leakage, but the method itself was shaped by this cohort; whether it carries over needs a new cohort.

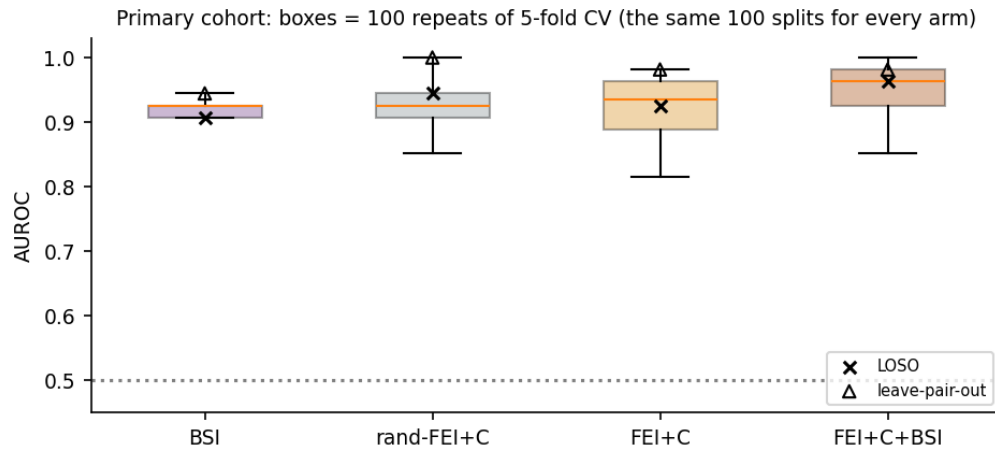


Figure 19: The results of Experiment 8 for BSI, FEI+GTJ, its random-init twin, and BSI+FEI+GTJ combined, under 100 repeats of 5-fold cross-validation on the primary stroke cohort ($\times$ = leave-one-subject-out, Δ = leave-pair-out, an even more stringent test that leaves out one stroke case and one control case every turn).

Table 15: Experiment 8: the FEI+GTJ is tested on the stroke input which has been spectrum corrected, and across three different cross validation schemes for robustness.

Arm	LOSO AUROC	5-fold $\times 100$ (mean $\pm$ sd)	Leave-pair-out	Permutation p (LOSO)
BSI (clinical baseline)	0.907	0.920 $\pm$ 0.012	0.944	0.005
rand-FEI+GTJ (random-init control)	0.944	0.930 $\pm$ 0.037	1.000	0.004
FEI+GTJ	0.926	0.918 $\pm$ 0.051	0.981	0.003
FEI+GTJ + BSI (combined)	0.963	0.951 $\pm$ 0.039	0.981	—

FEI+GTJ achieves 0.926 LOSO AUROC against the BSI's 0.907 ($\Delta = +0.019$, 95% CI $[-0.240, 0.280]$, $p = 0.523$); a similar point estimate that is not significantly different either way under all of the cross-validation schemes we considered. We didn't specify an equivalence margin beforehand, and an interval this wide cannot show equivalence either, so the cohort is just too small to rank the two. At the default 0.5 threshold, FEI+GTJ, BSI alone and FEI+GTJ+BSI combined all land on the identical confusion matrix (8/9 stroke cases and 6/6 controls correctly classified, one stroke case missed: Acc 0.933, BAcc 0.944, F1 0.941, Sens 0.889, Spec 1.000), differing only in AUROC, i.e. in how confidently each ranks that one missed case; the random-init twin is slightly worse at the same threshold (one extra false positive: Acc 0.867, BAcc 0.861, F1 0.889, Spec 0.833). It does not significantly outperform its own random-init twin on this task: FEI+GTJ vs. random-init twin is -0.019 LOSO AUROC ($p = 0.826$), not a significant difference in either direction. Splitting this by branch (exploratory, single arm, not pre-registered) shows the same pattern as TUAB: FEI's pre-training clears its random twin in 100% of repeated 5-fold splits, while GTJ's does not, only 16% of repeated splits. The single, deterministic LOSO split (every subject held out exactly once, so there is no fold-assignment randomness to repeat here) tells an even more dramatic branch story; FEI alone has a perfect 1.000 LOSO AUROC, with a threshold having no errors and 9/9 stroke / 6/6 control, while GTJ alone has 0.833 and the

random-init twins 0.907 (FEI) / 0.852 (GTJ). The 15-subject perfect score is still fragile, as one subject that was recorded atypically could flip it, and it is the repeated, randomized 5-fold win-rate above, and not this one-point estimate, that we would stand behind. FEI+GTJ combined with BSI (0.963 LOSO) is numerically superior to BSI alone in 82% of repeated splits; promising, but not statistically significant at 15 subjects. When the spectral-matching filter is refitted inside each fold instead (as in Table 19), FEI+GTJ remains at 0.926, and FEI+GTJ+BSI falls to 0.944, which is the same as its random-init twin. The two seeded pre-training re-runs of Experiment 5 were run through the same frozen pipeline: same spectrally matched input, no re-fitting, same encoder gates (only bootstrap comparisons were done, no permutation tests). FEI+GTJ scores 0.963 and 0.981 LOSO AUROC (original run 0.926; random-init twin 0.944, BSI 0.907), FEI alone 1.000 and 0.981 (twin 0.907) and GTJ alone 0.833 and 0.833 (twin 0.852); FEI's re-runs beat their twin in 100% and 96% of the repeated 5-fold splits, GTJ's in 16% and 15%. Against BSI, FEI+GTJ's re-runs differ by +0.056 ($p = 0.428$) and +0.074 ($p = 0.389$), neither significant. All three are evaluated on the same 15 subjects; the only source of variation in the spread is pre-training-seed, so they add no clinical sample, and the subject-level intervals above remain as wide as they were. The re-run GTJ encoders go through the same gates as the original, with $|z|$ at most 0.38 for the BatchNorm, embedding spread 0.74–0.82 of their spread on NMT, without any constant dimensions.

On this same cohort, the released ViT-M (VJEPA) achieves a LOSO AUROC of 0.981, which is higher than FEI+GTJ, FEI+GTJ+BSI, or BSI alone. But we do not read this as the released checkpoint working here. On this 15-subject cohort, it is just as collapsed as on TUAB (Table 9: std of tokens 0.0060, cosine of cross-recording 0.999994; each of its output features varies across the 15 subjects by only 0.0014 on average, against about 0.018 for a random-weight twin). A linear probe separates cases on this vanishing residual, which feature scaling blows back up, a residual that could just as easily track a recording-condition difference between stroke and control sessions as track stroke itself. Consistent with that, the same checkpoint's score drops from 0.981 on this cohort to 0.863 on the broader sensitivity cohort (a superset of these 15 subjects: 10 stroke vs. 8 controls, using the first 300 s instead of the 285 s rest block): the instability that a collapsed representation generates, not a true clinical signal.

5.3.4 Interpretation and Conclusions on Data

The released EEG-VJEPA checkpoints do not yield the same result as the paper, since they have been collapsed (Experiments 1-3). It's not corrected after fine-tuning the model for 500 epochs using the authors' recipe (Experiment 2), or by the simplest possible probe (Experiment 4). FEI+GTJ, built in response, is ahead of a classical baseline on the data it was trained on, although an untrained twin already accounts for about two-thirds of that margin (Experiment 5); on a different hospital's data, FEI's NMT-only pre-training still gives a significant gain (+0.048 AUROC averaged over five pre-training runs, 95% CI [0.015, 0.108]), and the fused model, re-pre-trained with that hospital's unlabelled recordings, remains close to the released checkpoint (difference not significant) and ahead of a classical baseline (Experiment 6), and, after a post-hoc input fix, is still close to a strong clinical baseline on a third structurally different task, too close for 15 subjects to separate (Experiments 7–8). A t-SNE/UMAP visual check, matching our original proposal's promised deliverable, shows that the two dimensional unsupervised layout is the wrong tool to view the reason that any of these encoders work, or why the released one doesn't.

The same per-recording predictions from the experiments in the above were re-examined outside the original proposal during a second analysis pass, without retraining. The question: when FEI and GTJ are off, are they off in the same way or in different ways, and does that account for fusion's successes and failures?

- Children under 18 account for 43% of all recordings on NMT, but 65% of all abnormal-labelled ones, and FEI labels 62% of healthy 0-6 year olds as abnormal: by far its single largest source of false alarms, despite FEI's best performance being on adults (age alone predicts the label with AUROC 0.609 on NMT, and within the normal class, FEI's scores correlate with age at $r = -0.42$). The primary mode of failure of FEI is age dependent. Our working, untested interpretation is that FEI's pretext task (predicting how its target encoder embeds a window once random frequency bins are removed) makes it sensitive to slow-wave activity that is a natural part of a healthy developing brain in young children, but not a sign of disease. This pattern can not be checked on TUAB as there are only 2 recordings under 18.

- FEI is also weaker on adult females, but only a flag. On NMT adults, FEI reaches 0.773 AUROC on women

vs. 0.864 on men; the gap narrows for GTJ (0.832 vs. 0.852) and for FEI+GTJ (0.846 vs. 0.883). This is the same direction that is marked on TUAB (FEI 0.865 on women, 0.883 on men). This is not just the age effect rewritten differently as it also exists in children only (FEI 0.756 female vs. 0.767 male), but it is not a finding we have tested or found a mechanism for – it was an observation for future work.

- On NMT, the per-recording errors of FEI and GTJ overlap only slightly (score correlation 0.60), and fusion repairs 321 of the 622 errors on FEI while introducing only 21 new ones. With the NMT-only encoders on TUAB, on the other hand, the two branches' scores correlate at 0.84, and 44 of FEI's 58 errors are also GTJ errors, so fusion has little left to fix: this NMT-only FEI+GTJ (0.857) is a bit lower than FEI alone (0.872; paired difference -0.015 , 95% CI $[-0.047, +0.016]$, $p = 0.36$). The fused row of Table 12 (0.862) is the "joint" version that has been separately re-pre-trained. The two branches also have different failure reasons: FEI's primary failure is age (62% of healthy 0-6 year olds on NMT are flagged as abnormal by FEI), while GTJ's primary failures are related to the distribution-shift, and not to age per se. Now, fusion is only beneficial when its two inputs disagree, and this is why Experiment 5 has high fusion gain on the NMT and Experiment 6 has less gain on the TUAB.

- When it comes to being confidently wrong, GTJ is so about twice as often as FEI. Considering only each model's mistakes, FEI's mistakes tend to be close to the 0.5 decision boundary, a clinically useful property - that is, when a case is uncertain, it could be flagged for a human expert to review; GTJ makes confident, big mistakes about twice as often on NMT and more than seven times as often on TUAB. Because fusion carries over GTJ's confident errors, this is a calibration weakness unique to GTJ, separate from its overall ranking ability.

- A single global decision threshold is no good for different ages. At the default threshold of 0.5, FEI+GTJ succeeds in detecting only 50–69% of the abnormal recordings in adults while marking large proportions of healthy children's recordings as abnormal, since the classifier is trained on a set of examples in which most abnormal examples are children, and hence its classification boundary is implicitly tuned for them. The ranking (AUROC) of the model is good for adults; what fails is the single shared threshold.

- In an exploratory run, when pre-trained on NMT plus unlabelled TUAB, instead of NMT alone, GTJ's NMT AUROC decreased in all 5 folds (-0.014) and its embedding spread on NMT also decreased, indicating that GTJ shifts its representational capacity to the domain it is exposed to instead of learning a more general structure. In this configuration, more data fed into GTJ during pre-training was detrimental to the performance. Consistent with this, GTJ's past-predicts-future pretext loss plateaued after about 10 epochs, indicating that for this architecture and this amount of data, the task is simply too easy to force deep, transferable learning.

- One Fig-3-head number doesn't seem to be reproducible even by us, on the same seed. Maruf re-ran the attention-head extraction behind the Fig-3 attentive probe (Figure 15) to double-check it before freezing the plots, and with the same seed, the same code and the same checkpoint, got 72.1 AUROC the first time, and 68.7 on the re-run (Table 7 reports the re-run; with the first run its five-subset mean would read 0.720). This is likely due to the non-deterministic nature of one of the attention kernels at the GPU level (by default cuDNN does not guarantee bit-identical output across runs unless forced to run deterministic algorithms, which we weren't). We did not chase this down further; it is flagged here because it is a real gap in our own reproducibility as well, beyond the paper's. The numbers in §4.2 AttentionClassifier are bit-identical over re-runs and the exact value of the Fig-3 head is irrelevant for any conclusion in this report.

The checkpoints released with EEG-VJEPa are collapsed and will not reproduce the paper, but our token-spread diagnostic is repeatable and the released training config provides a likely cause. FEI+GTJ beats our classical baseline on NMT, with pre-training adding a consistent but not significant $+0.027$ AUROC over an untrained twin (three-run mean 0.861 vs 0.834); on stroke, after a post hoc input fix, its AUROC comes close to BSI's, with too few subjects to consider the two equivalent. The gain ($+0.048$ AUROC averaged across five pre-training runs, 95% CI $[0.015, 0.108]$) of FEI's NMT-only pre-training is significant across hospitals, and the fused model, which is re-pre-trained using TUAB's unlabelled recordings, is not significantly different from the released ViT-M there. On stroke, only FEI demonstrates a pre-training gain, and no model is

significantly better than BSI. These results are compared to the original proposal in Table 16.

Table 16: Act 0 — original proposal targets what has been done.

Proposal aim	Status	Evidence
Part 1: reproduce the paper's frozen TUAB result	Done faithfully, as far as the paper's text specifies; result not reproduced	43/48 files identical to upstream; 0.671–0.762 vs. paper 0.877–0.879
Explain the shortfall	Achieved	Collapse: token std 0.005 vs. 0.63 for random weights (Exp. 3)
Added: fine-tune the released ViT-M	Done; no gain from pre-training	0.755 ± 0.053 vs. 0.768 ± 0.114 from scratch, $p = 0.81$ (Exp. 2)
Changed: FEI+GTJ as a working pre-trained encoder	Achieved; pre-training gain modest	original run 0.865 ± 0.024 on NMT (three-run mean 0.861) vs. band power 0.786 and untrained twin 0.834 (pre-training +0.027, not significant); on TUAB, FEI's NMT-only pre-training +0.048 averaged over five runs (95% CI [0.015, 0.108]; original run +0.059); fused, with TUAB's unlabelled recordings added, close to released ViT-M (not significant) and beats band power (Exp. 5–6)
Part 2: healthy vs. stroke, frozen encoders + BSI, LOSO	Done	Gate passed $p = 0.003$; BSI 0.907; FEI+GTJ 0.926 (seed re-runs 0.963, 0.981) after a post-hoc input fix, not separable from BSI (Exp. 7–8)
Deliverable: t-SNE / UMAP of learned features	Done	No class clusters in any map, learned or random (10-NN agreement ≈ 0.60 vs. chance 0.502)
Our own EEG-VJEPA pre-training from scratch	Dropped	Out of scope by team decision, in favour of the diagnosis-and-replacement work above

5.4 Limitations of Tools

- **Compute.** All of this report, from pre-training to evaluation, was performed on a single consumer GPU (RTX 5060 Ti) with a thermal guard to pause or even kill a job if the CPU is too hot. We couldn't afford many random-init seeds (3, not 10) for some controls, and the much larger pre-training runs used by the released EEG-VJEPA or by CBraMod/LaBraM were out of reach. Checks that we are not able to run, and why, are listed in Appendix §11.5.

- **Cohort size.** There are 15 subjects in the stroke cohort. All confidence intervals are broad and some (such as FEI+GTJ versus its twin with random starting conditions) are not restrictive enough to exclude even fairly large differences in either direction.

- **Demographic confound of the stroke cohort.** According to the dataset's own record description [13], the stroke patients are 55.9 ± 14.2 years old, 6 men and 4 women, and the controls are 46.3 ± 23.6 years old, 1 man and 6 women; these figures are for the whole record, not exactly the 15 that we are analysing. Encoders can read age and sex off EEG, so one possible explanation for part of any stroke-vs-control score in Experiments 7–8 is that the score was less a reflection of stroke than the reading of the age and sex difference. With 15 subjects, there is no way to separate the two statistically: Soumya flagged this when making the cohort table and we checked that there is no way to get this balance by just re-splitting the cohort, as the imbalance is with regard to who was recruited, not how we split them.

- Data access. The copy of TUAB we're using came from an unofficial mirror, not from the licensed channel (see Legal Issues, §5.5.3). That restricts the things we can safely do with those numbers.

- Information that is not contained within the reference paper. The EEG-VJEPA paper doesn't specify which 546 recordings are in each of the five reported training subsets, from what training epoch do the headline numbers in the paper come from, or what fine-tuning recipe was used to get the 0.885 AUROC. We seeded our own choices in a seeded sampler (code/tuab_released_eval.py – the 5 546-recording subsets are listed file by file in docs/act0_repro/tuab_subsets.csv), but it is not guaranteed that these choices are the same as the authors'.

5.5 Impact Assessment

5.5.1 Assessment of Societal and Cultural Issues

In our TUAB result (Experiment 6), our self-supervised encoder is pre-trained on a lower resource-setting EEG dataset (FEI, with pre-training on NMT) and yet transfers a measurable benefit to another hospital's data without seeing any of its data in pre-training (the linear probe on top is still trained on the other hospital labels).

5.5.2 Assessment of Health and Safety Issues

FEI's false alarm rate in young children (§5.2.1) and the other demographic-specific failure modes (§5.3.4) indicate that for the time being FEI+GTJ should stay a research artefact and not be used clinically as a decision support tool: not without age-stratified validation and recalibration first.

5.5.3 Assessment of Legal Issues

The official TUH Abnormal EEG Corpus is provided by Temple University under a Data Use Agreement (DUA), signed with the maintainers of the corpus (NEDC) before access is granted. The copy we used was from an unofficial, publicly re-hosted Kaggle mirror: it includes the corpus's own v3.0.1 README, its directory structure and file counts (1,371 normal; 1,346 abnormal training recordings; 150 and 126 evaluation) correspond to those in the README, but we have no DUA-covered official copy to compare against, so we cannot verify that the copy is byte-for-byte identical to the licensed copy (the SHA-256 of all the files used are in docs/act0_repro/raw_recordings_sha256.csv, Table 21). It is not covered by that licence. All numbers derived from the TUAB in this report (Experiments 1, 2, 3, 4, 6, and the visual check) are presented for internal academic evaluation within this course submission only, and should not be published, re-distributed or reused anywhere outside it. If the public use of the TUAB numbers is desired at some later time (in a paper, public leaderboard entry, public repository result), then it must be done with a properly licensed copy of TUAB, via the official DUA process; alternatively, it must eliminate the TUAB numbers entirely and depend solely on NMT and the stroke dataset that is openly licensed.

5.6 Sustainability Evaluation

Moreover, small models and single-GPU training also limit compute demand (§5.2.2). In fact, our two encoders have 1.15M (FEI) and 1.88M (GTJ) parameters (including online and target copies; Tables 1–2), and were trained for 60 epochs on a few thousand recordings, whereas CBraMod and LaBraM [5, 6] pre-train on tens of thousands of hours. It still clears a classical baseline, and with the addition of pre-training with unlabelled recordings from TUAB, the score is close to a released large-scale checkpoint's linear-probe score on TUAB, with no significant difference. If that holds more broadly, EEG models could be sized to fit the data that can be gathered in a clinical setting, which is often small and costly to collect, rather than defaulting to the

largest model on the shelf.

5.7 Ethical Issues

All three datasets were gathered, anonymised and published by the primary source re: ethics and consent and we are secondary users of already published, anonymous data with no data collection. We revealed that our TUAB is from an unofficial source (§5.5.3) limited our numbers to this internal submission and said what proper licensing would require. If someone were to use it as a clinical tool instead of a term paper, a model with an unrevealed failure mode (FEI over-flagging healthy young children at six times that of adults) could do real damage.

6 Reflection on Individual and Team work

6.1 Individual Contribution of Each Member

Ahmed Mashfiqur Rahman (2106114) led the model architecture: studying V-JEPA and ViT-M, and the design and building of both FEI and GTJ, including the attention-pooling and linear classifier heads and the shape/parameter checks that validated them.

Soumya Das Prottay (2106116) performed preprocessing for all three datasets utilized in this report (NMT, TUAB, stroke cohort), the same preprocessing pipeline used to reproduce the recipe in §4.1 of the paper, data quality checks for each of the three sets, windowing and training subset construction for each of the three sets, and the stroke spectrum-matching fix (Experiment 8), as well as the BSI feature extraction for the clinical baseline (Experiment 7).

The training and results: running the pretraining and fine-tuning jobs, reproducing the released checkpoint and diagnosing its representation collapse (Experiments 1–4), the cross-validation, the statistical significance testing behind each number reported, the tables and figures, and the failure-axis analysis in §5.3.4, were led by Maruf Hasan Mahi (2106121).

In addition to these individual leads, all three members of the team participated in all three of the following team activities: joint design discussions (as described below under Mode of Teamwork); experiment design decisions (as described below under Mode of Teamwork); and review of all results prior to their inclusion in this report.

6.2 Mode of TeamWork

Together we made major design decisions such as what failure mode to explore next, what experiment a result would need to have in order to be trusted and what to exclude from scope. Each member performed her/his part separately. Before accepting any result, we discussed them together.

6.3 Diversity Statement of Team

Group 01 combined interests in signal processing, machine learning and clinical applications. That combination gave rise to the project: we didn't consider EEG to be a generic input for a black box model, but a signal with its structure.

6.4 Log Book of Project Implementation

In week 5 of term (Level 4, Term I) our proposal was accepted. Work was carried on through week 7, then a one-week mid-term break, followed by work in weeks 8-14, when the report and demonstration were completed.

Table 17: Log book Act-0 Project implementation week 5-14.

Week	Milestone achieved	Individual role	Team role
Week 5	Proposal accepted; literature review of EEG-VJEPa, the JEPa paradigm, and existing EEG foundation models	Reading, initial pipeline scoping	Joint proposal writing
Week 6	Hardware thermal-guard set up; NMT dataset acquired; paper's §4.1 preprocessing pipeline reproduced	Preprocessing implementation	Design discussion, review
Week 7	Released ViT-M/ViT-B checkpoints evaluated on NMT-matched input; representation collapse first suspected	Evaluation pipeline, diagnostics	Joint result review
Mid-term break	FEI designed and implemented from scratch; validated with unit self-tests	FEI implementation and testing	Design discussion
Week 8	GTJ designed and implemented; both branches unit-tested individually before any fusion	GTJ implementation and testing	Design discussion, review
Week 9	FEI+GTJ fusion built; 5-fold NMT cross-validation run against the band-power baseline (Exp. 5)	Fusion + CV pipeline	Joint result review
Week 10	TUAB dataset acquired and preprocessed; released-checkpoint and FEI+GTJ transfer evaluation begun (Exp. 1, 6)	TUAB pipeline, transfer evaluation	Joint result review
Week 11	Collapse formally diagnosed (Exp. 3); fine-tune-vs-scratch (Exp. 2) and simplest-probe (Exp. 4) experiments run	Diagnostics, statistics	Design discussion, review
Week 12	Stroke dataset acquired; Brain Symmetry Index baseline implemented; pipeline-soundness gate passed (Exp. 7)	BSI implementation, gating	Joint result review

Week	Milestone achieved	Individual role	Team role
Week 13	GTJ's cross-dataset BatchNorm collapse on stroke diagnosed, traced to a silently wrong assumption that the stroke recordings used NMT's own filter band; spectrum-matching fix designed, gated, and applied (Exp. 8)	Fix design and implementation	Design discussion, review
Week 14	Failure-axis analysis completed; all results, figures and this report finalised for demonstration	Failure-axis analysis, report writing	Joint review and rehearsal

7 Communication to External Stakeholders

7.1 Executive Summary

A recently published AI model, called EEG-VJEPa, claimed to use a method for understanding videos to differentiate healthy brain recordings from abnormal ones. We tested this claim and discovered that the released version isn't really functional as advertised: it produces virtually identical answers for all patients. We could demonstrate this directly. Instead we constructed two small models—one for the frequency content and another for how the rhythm of a signal changes over time. Together they outperform a simple classical measurement on the dataset they were trained on (an untrained copy of the same design already does most of that) and come close to an existing clinical measurement, with too few patients to distinguish the two, on a third task: distinguishing stroke patients from healthy volunteers. On a different hospital's data, the frequency model's training was still useful on its own, across five training runs. Training did not carry over for the rhythm model, and retraining both models with the unlabelled recordings of that hospital changed little: either way the combined model scored close to the released model and not measurably better than an untrained copy of itself.

7.2 Reproducing Our Results

The code, saved model checkpoints and saved per-recording predictions for every number, table, and figure in this report are packaged into a set of scripts, one for each experiment (Table 5). Table 4 shows the script behind each experiment's input. Using those inputs, pre-train an encoder (``fei_pretrain.py`` or ``fei_branchC.py``, both with a ``--selftest`` flag for a fast sanity check with no real data, and a ``--cv`` flag for the full five-fold cross-validation reported here), score it with ``feic_nmt_anchor.py`` (Experiment 5) or ``tuab_feic.py`` (Experiment 6), and evaluate a frozen or fine-tuned released checkpoint with the corresponding ``tuab_*_eval.py`` script. The code is in github.com/mashfiquurrahman2020-coder/eeg-jepa-act-0, a repository with results computed on our unofficial copy of TUAB (§5.5.3); course examiners can inspect the exact commit below. Its `docs/act0_repro/` folder holds the fold, subset and cohort lists, the SHA-256 of every checkpoint, and the pinned environment (Appendix §11). The code, checkpoints and results used for this report are commit `1594b90b5f61e992f6529dca47673d7a83857190`: `git clone https://github.com/mashfiquurrahman2020-coder/eeg-jepa-act-0.git`, then `git checkout 1594b90b5f61`, then `pip install -r docs/act0_repro/requirements-act0.txt` under Python 3.9. Appendix §11.6 contains the pre-training commands and §11.9 the data used for each experiment. There is no need to download anything besides the data and the released EEG-VJEPa weights, since all 70 FEI/GTJ checkpoints are committed in the repository. There are two types of reproduction that follow. The saved checkpoints and predictions reproduce every reported number exactly. Re-training does not: the original FEI and GTJ runs do not set a random seed so cannot be reproduced bit by bit, and cuDNN kernels were left non-deterministic, so even the seeded re-runs should be expected to reproduce results statistically, not bit by bit.

8 Project Management and Cost Analysis

8.1 Computational Resources Used

This software project has no actual BOM. The resource budget used was compute. No paid cloud compute was utilized for all pre-training and evaluation performed on a locally owned workstation with one consumer grade GPU (NVIDIA RTX 5060 Ti). The direct monetary cost was for the electricity powering the workstation. For this project, the practical cost was the time and amount of thermal headroom: a shared hardware guard suspended training if temperatures exceeded a safe threshold on the CPU, thus limiting the number of experiments (and control seeds with random initializations) we could accommodate in the term.

To give a concrete number: ``code/runs/act0/feic_joint.log`` records the 5-fold cross-validation run that pre-trains FEI (1.15M parameters, Table 1) and GTJ (1.88M parameters, Table 2) separately, not against a common loss, on the joint NMT+TUAB training pool, the "joint-pretraining" data variant this report compares against its main NMT-only numbers. It trains 10 encoders (60 epochs of 145 steps each, one FEI and one GTJ per fold). The total job takes about 78 minutes of GPU time.

8.2 Timeline of Project Implementation

Refer to Table 17 (§6.4) for the detailed week by week log book. In summary: weeks 5-7, proposal, lit review, data acquisition, and first (unsuccessful) reproduction attempt; mid-term break and week 8, building FEI and GTJ from scratch, each unit-tested independently; weeks 9-11, NMT validation, TUAB transfer, formally diagnosing representation collapse; weeks 12-13, the stroke cohort, the clinical BSI baseline and the spectrum-matching fix; week 14, failure-axis analysis and this report.

9 Future Work

- Age-aware modelling. The most obvious practical conclusion from this report is that the false-alarm rate of FEI is much higher in healthy young children than in adults (§5.3.4). An age-conditioned probe should follow, with AUROC routinely reported by age band.
- General patch for GTJ. The spectrum-matching fix in Experiment 8 is built for the stroke cohort specifically; a dataset-agnostic one (such as replacing BatchNorm with per-recording normalisation, for example) could allow GTJ to transfer the way FEI currently does. The same work ought to cover GTJ's confidently wrong errors (§5.3.4) as well: a calibration step — e.g., temperature scaling fit on a held-out fold — prior to trying to interpret its output as a probability, not a ranking. Before any of that, the recipe of spectral matching must be frozen and used unchanged on a new stroke cohort: Experiment 8 only demonstrates how it repairs the cohort it was developed on.
- A correctly licensed TUAB assessment. The obtaining of the official, DUA-covered TUH Abnormal EEG Corpus would enable the findings derived from the TUAB to be verified, extended and published without the limitation in §5.5.3 which is internal only.
- Motor imagery. Separate extension of this work into motor-imagery brain-computer interfaces, not covered in this report, has already begun by the team, following GTJ's design based on spectral-dynamics.

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11 Appendix: Reproducibility Records

11.1 Environment and Checkpoint Provenance

Table 18: Software environment, upstream code and released-checkpoint provenance (docs/act0_repro/environment.txt and checkpoints_sha256.csv).

Item	Value
Python	3.9.25
PyTorch	2.8.0+cu128 (CUDA 12.8, cuDNN 9.10.2)
Key packages	MNE 1.8.0, scikit-learn 1.5.2, SciPy 1.13.1, NumPy 1.26.4, pandas 2.3.3 (all 111 pins in docs/act0_repro/requirements-act0.txt)
GPU	NVIDIA GeForce RTX 5060 Ti
Upstream EEG-VJEPa code	github.com/amir-hojjati/eeg-vjepa, commit c739fad5ad57ec4439a094343901a245edf611ec
Released checkpoints	Hugging Face amir-hlp/EEG-VJEPa, commit ec98405f5f3423fd59d3296157ee1570efd8d947
ViT-M (4×30×4) SHA-256	5fee4a6c08123cfb2d18d86e5d483aac32630ceed7ed1b2babc052799c48db79
ViT-B (4×30×2) SHA-256	66e9673f8ce42c3c973470821777a793a37db2ddb58dfd040a664560f7eefc85
Our 70 FEI/GTJ checkpoints	size and SHA-256 of each in docs/act0_repro/checkpoints_sha256.csv

11.2 Stroke Cohort: Pre-registration, Post-hoc Input and Fold-local Check

The stroke analysis plan was written down before any LOSO run, in a file kept in our project repository (runs/act0/phaseB/PREREGISTRATION.md, dated 2026-09-25). Original results record its SHA-256. The current script allows runs without this file and records null if absent. Because this file is in our own repository rather than a public registry, it shows the order of our work but cannot prove it to an outsider. Under the plan, FEI+GTJ scored 0.833 LOSO AUROC, below its random-init twin (0.889) and BSI (0.907).

The spectral-matching input for Experiment 8 was designed after that result: its recipe, gates and comparisons were written to runs/act0/phaseB_clean/POSTHOC.md on 2026-09-26 prior to any LOSO run on that input, so all the numbers in Experiment 8 are post-hoc in that sense. Its filter is fitted once, without any labels, on all 15 subjects pooled, which means each held-out subject influences the filter applied to it. To ensure that this does not drive the result, stroke_phaseB_foldlocal.py refits the filter in each of the LOSO folds on the 14 training

subjects only, and applies it unchanged to the held-out one. A comparison of the two is shown in Table 19. The pre-registration also fixes both cohorts. For each of the subjects, the primary cohort takes the same 285 s of pure rest prior to any task marker (the shortest rest block, PAC05's, is 285.0 s); stroke PAC02–PAC10 (9) against the dataset authors' own six controls. PAC01 and controls 02 and C01 are excluded because they have at most 52 s of rest before the task. The sensitivity cohort follows the authors' definition, the first 300 s of each file (10 stroke, 8 controls); for PAC01, 02 and C01 that window includes task data, and it is therefore reported as a sensitivity check only.

Table 19: Stroke primary cohort, LOSO AUROC with the spectral-matching filter fitted on all 15 subjects pooled (Experiment 8) or refitted inside each fold on the 14 training subjects. The BSI alone does not apply any filter (0.907).

Arm	Pooled filter (Experiment 8)	Fold-local filter
FEI+GTJ	0.926	0.926
FEI+GTJ+BSI	0.963	0.944
rand-FEI+GTJ	0.944	0.944
rand-FEI+GTJ+BSI	0.944	0.944
FEI	1.000	1.000
GTJ	0.833	0.833
rand-FEI	0.907	0.907
rand-GTJ	0.852	0.852

Only the combined FEI+GTJ+BSI arm moves (0.963 to 0.944), which removes its small numerical advantage over its random-init twin; FEI+GTJ itself is unchanged, as is its AUROC close to BSI's that Experiment 8 is based on.

11.3 Experiment 5 Corrected Statistics

Table 20: NMT, Experiment 5 – all 11 comparisons. Pre-trained arms are per-fold means over three pre-training runs, and random-init arms are over five seeds. Corrected p is the Nadeau–Bengio resampled t-test [20] ($\text{variance} \times (1/5 + n_{\text{test}}/n_{\text{train}})$, $n_{\text{train}} \approx 1,934$, $n_{\text{test}} \approx 483$, 4 degrees of freedom), and Holm [21] corrects over all eleven. The ordinary paired t-test over folds (naive p) is provided for comparison only. Averaging the three runs within each fold before the test folds training-run variation into the fold means; a hierarchical model would estimate the two sources separately.

A vs B	Mean A	Mean B	Δ AUROC	Folds A > B	Naive p	Corrected p	Holm p
FEI vs band power	0.820	0.786	+0.034	4/5	0.038	0.112	0.447
GTJ vs band power	0.831	0.786	+0.045	5/5	< 0.001	0.003	0.027
FEI+GTJ vs band power	0.861	0.786	+0.075	5/5	< 0.001	0.004	0.029
rand-FEI vs band power	0.810	0.786	+0.024	4/5	0.109	0.243	0.486
rand-GTJ vs band power	0.794	0.786	+0.009	5/5	0.027	0.086	0.431
rand-FEI+GTJ vs band power	0.834	0.786	+0.048	5/5	< 0.001	0.002	0.026
FEI vs rand-FEI	0.820	0.810	+0.010	3/5	0.179	0.338	0.486
GTJ vs rand-GTJ	0.831	0.794	+0.036	5/5	0.005	0.019	0.116
FEI+GTJ vs rand-FEI+GTJ	0.861	0.834	+0.027	5/5	0.055	0.149	0.447
FEI+GTJ vs FEI	0.861	0.820	+0.041	5/5	< 0.001	0.003	0.027
FEI+GTJ vs GTJ	0.861	0.831	+0.030	5/5	0.003	0.012	0.082

11.4 Reproducibility Files

Table 21: Files in docs/act0_repro/ (written by code/act0_repro_manifest.py). The raw recordings are not redistributed: NMT [7] and stroke cohort [13] are public, and TUAB must be obtained from its maintainers under their data use agreement (§5.5.3). Their per-file SHA-256 are provided instead, which allows any copy to be checked, byte for byte.

File	Rows	Contents
nmt_folds_seed0.csv	2,417	Every NMT recording's label and test fold (Experiment 5)
tuab_subsets.csv	2,730	The five 546-recording TUAB training subsets s0–s4: file and subject
stroke_cohorts.csv	33	Primary (15) and sensitivity (18) cohort membership, label and rest seconds
checkpoints_sha256.csv	72	Size and SHA-256 of the released checkpoints and every FEI/GTJ checkpoint
requirements-act0.txt	111	Pinned Python packages
environment.txt	6	Table 18's environment lines
raw_recordings_sha256.csv	5,501	Dataset, path, size and SHA-256 of every file in the three raw-data folders: TUAB's 2,993 EDFs; NMT's 2,417 EDFs and its label file; the stroke folder's 90 files (code/act0_raw_hashes.py)

11.5 Checks Not Run

Table 22: Checks which would enhance this report, but were not performed, and why not, and what the report provides.

Check	Why it was not run	What the report gives instead
Re-running EEG-VJEP A pre-training with reg_coeff restored (Experiment 3)	The released config (cluster_vit16_EEG.yaml) pre-trains for 400 epochs. One run of that length, let alone the repeats a causal test needs, did not fit the GPU time we had.	The collapse itself is measured (Table 9); reg_coeff = 0 and the lowest-loss checkpoint (epoch 353) are given as hypotheses for its cause.
Nested cross-validation for GTJ's window length (Experiment 5)	The 5/20/40 s comparison would have to be repeated inside every outer fold, multiplying GTJ pre-training by the number of inner folds.	The selection is disclosed with the sweep, and GTJ beats band power at every window length (0.820, 0.836, 0.823).
NMT-only GTJ and FEI+GTJ re-runs on TUAB (Experiment 6)	The TUAB re-runs went to FEI, whose NMT-only gain is the claim the report makes; NMT-only GTJ already scores below its random-init twin, and no conclusion rests on it.	NMT-only GTJ and FEI+GTJ from the original run against their twins (Table 12); FEI over five runs and the adapted encoders over three (§5.3.3).
Permutation tests for Experiment 8's pre-training re-runs	Each needs 1,000 full LOSO re-runs per arm and run, the slowest part of Experiment 8.	Paired bootstrap comparisons of each re-run with the same random-init twin and with BSI over the 15 subjects (§5.3.3).
Five random-init seeds for the TUAB and stroke twins	Not run in the time left before submission; the three-seed twin ensemble is the reference behind every reported TUAB and stroke pre-training gain.	Experiment 6's multi-run bootstrap resamples the three seeds as a level of its own (Table 24), so their variation enters the interval, if coarsely.
Repeated, deterministic runs of the Fig-3 attentive head (Experiment 1)	No conclusion depends on that head's exact value; the argument uses the linear probe and the §4.2 head, which is bit-identical across re-runs.	Both runs are reported (72.1 and 68.7) and Table 7 uses the re-run.
Evaluation on the licensed TUAB release	Access needs a data use agreement with TUAB's maintainers (§5.5.3), which we did not have.	TUAB numbers are restricted to this submission (§5.5.3); a licensed evaluation is listed as future work.

Check	Why it was not run	What the report gives instead
An equivalence test between FEI+GTJ and BSI on stroke (Experiment 8)	No equivalence margin was set in advance, and with 15 subjects the 95% CI of the difference (−0.240 to 0.280) is far wider than any useful margin.	The two are reported as not separable, not as equivalent.
The spectral-matching recipe, frozen, on a new stroke cohort (Experiment 8)	We had no second stroke EEG cohort within the project.	Experiment 8 is reported as a post-hoc repair on the cohort that prompted it; external validation is future work.

11.6 Pre-training Configuration and Commands

All encoders pre-trained for this report are based on the following recipe. All runs are done with no learning-rate schedule, mixed precision, gradient-clipping, or PyTorch's deterministic-algorithm flags, and without the selection of checkpoints, keeping the final (60th) epoch weights. The cuDNN kernels were left non-deterministic, so even re-running the seeded runs will not be bit-identical. The random-init twins are built from the same architectures with torch seed = twin index (seeds 0–2 on TUAB and stroke, 0–4 on NMT); the multi-run bootstrap of Experiment 6 resamples the three TUAB seeds (Table 24).

Table 23: For all encoders in this report, pre training data, seeds, recipe and command were used. All share AdamW (learning rate 2×10^{-4} , weight decay 10^{-4}), batch 128, 60 epochs, 4 random windows per recording per epoch, d = 256, h = 128 and target EMA 0.996.

Encoders	Pre-training data and seeds	Window and branch settings	Command and checkpoints
FEI, NMT only (Experiments 5, 6, 8)	The NMT training folds of the seed-0 5-fold split (Table 21); no seed set (the original runs are unseeded)	L = 1,000 samples (5 s at 200 Hz)	python code/fei_pretrain.py --cv → fei_enc_cv_s0_f{k}.pt
GTJ, NMT only (Experiments 5, 8)	As for FEI; no seed set	L = 4,000 (20 s), STFT n_fft 200, hop 100, context 60%, GRU	python code/fei_branchC.py --cv --L 4000 --hop 100 --tag h20 → branchC_h20_enc_cv_s0_f{k}.pt
FEI and GTJ replicates 1 and 2 (Experiment 5, Table 20; FEI also in Experiment 6, both in Experiment 8)	As above; torch, NumPy and Python random seeded with 100 × replicate + fold	As above	python code/act0_nmt_pretrain_replicate.py
FEI replicates 3 and 4 (Experiment 6 only)	As for replicates 1 and 2	As for FEI	python code/tuab_fei_reps.py → fei_rep{3,4}_enc_cv_s0_f{k}.pt

Encoders	Pre-training data and seeds	Window and branch settings	Command and checkpoints
FEI and GTJ, joint NMT+TUAB (Experiment 6)	NMT training fold k plus TUAB's 2,717 unlabelled training recordings; torch seed = k	As above	python code/feic_joint.py → fei_joint_enc_cv_s0_f{k}.pt, branchC_h20_joint_enc_cv_s0_f{k}.pt
FEI and GTJ, joint NMT+TUAB, replicates 1 and 2 (Experiment 6)	As for the joint encoders; torch, NumPy and Python random seeded with $100 \times \text{replicate} + \text{fold}$	As above	python code/tuab_feic_joint_reps.py → fei_joint_rep{r}_enc_cv_s0_f{k}.pt, branchC_h20_joint_rep{r}_enc_cv_s0_f{k}.pt

11.7 Statistical Procedures

Table 24: The statistical test used for each type of comparison in this report.

Comparison	Procedure	Where used
Confidence interval of one model's AUROC or BAcc	Percentile bootstrap over test recordings (subjects on stroke), 2,000 resamples, seed 0, 2.5th–97.5th percentiles (code/act0_metrics.py)	Experiments 4, 6, 8
Two models on the same test set (Δ AUROC)	Paired bootstrap: both models scored on the same 2,000 resamples; CI from the percentiles of Δ , and p = share of resamples with $\Delta \leq 0$ (one-sided)	Experiments 6 and 8 (e.g. FEI+GTJ – ViT-M on TUAB, $p = 0.147$)
Mean pre-training gain over several pre-training runs (Δ AUROC)	Multilevel bootstrap, 2,000 resamples, seed 0: each resample draws, with replacement, the pre-training runs, the random-init twin's three seeds (re-ensembled) and the test recordings; Δ = mean AUROC of the drawn runs minus the drawn twin ensemble's AUROC; CI from the percentiles of Δ , p = share with $\Delta \leq 0$ (pooled) in code/tuab_fei_reps.py). Against the released ViT-M, one fixed model, only runs and recordings are drawn.	Experiment 6 (five NMT-only FEI runs; three adapted runs)
Pre-training re-runs on the stroke cohort	Each run scored on its own with the paired bootstrap over the same 15 subjects; runs are not pooled, so their spread (seed variation) stays separate from subject-level uncertainty.	Experiment 8
Released vs random-weight ViT-M	One-sided exact Mann–Whitney U on per-run final AUROCs: 5 released-checkpoint runs (one head per training subset, s0–s4) against 3 random-weight runs (s0–s2, each with its own random weights, torch seed = subset number). Alternative: released below random in Experiment 1, above random in Experiment 4. With 5 and 3 runs the smallest possible p is $1/56 = 0.018$; two-sided p -values are twice the one-sided ones.	Experiments 1 and 4
Fine-tuned vs from-scratch ViT-M on the same five training subsets	Two-sided paired t-test and Wilcoxon signed-rank over the 5 subsets. Any two subsets share 98–134 of their 546 recordings, which both tests ignore, so their p -values lean small; at $p = 0.81$ that cannot change the conclusion.	Experiment 2
Two arms on the same NMT folds	Paired t-test over the 5 folds; Appendix §11.3 repeats all eleven with the Nadeau–Bengio corrected t-test and Holm's adjustment	Experiment 5 (Table 20)

Comparison	Procedure	Where used
Stroke: one arm against chance	Label permutation, the full LOSO re-run for each of 1,000 shuffles; $p = (1 + \# \text{null} \geq \text{observed}) / 1,001$	Experiments 7 and 8
Stroke: robustness to the split	100 repeats of stratified 5-fold with the same splits for every arm (win rate = share of repeats arm A beats arm B; permutation p from 200 shuffles $\times$ 10 repeats); leave-pair-out with a stratified 2,000-resample bootstrap over subjects	Experiment 8 (Table 15, Figure 19)
Stroke: classical features, stroke vs control	Two-sided Mann–Whitney U; the pre-declared gate is broadband BSI at $p < 0.05$	Experiment 7 (Table 13)

Only the eleven NMT comparisons of Experiment 5 are adjusted for multiplicity (Holm, Table 20). The stroke pre-registration fixed the gate and three paired comparisons, all involving the released ViT-M (ViT-M+BSI vs BSI, ViT-M vs its random twin, FEI+GTJ vs ViT-M), on the pre-registered input, while the comparisons of Experiment 8 on the spectrum-matched input were fixed in POSTHOC.md before any LOSO run was conducted on it, but after the pre-registration. All other p-values reported in this report are exploratory, unadjusted: evidence regarding that one comparison – not a claim about the entire family.

11.8 Stroke Spectral-Matching Filter

The input for Experiment 8 is created using the code/stroke_phaseB_clean.py, following runs/act0/phaseB_clean/POSTHOC.md, which was written prior to any LOSO runs with this input. Each recording gets the dataset authors' bad-channel interpolation, a common average reference, the 19 NMT channels (T7/T8/P7/P8 renamed T3/T4/T5/T6), resampling to 200 Hz with no band-pass, and a per-channel z-score. The target is the mean log10 Welch spectrum of 100 random, fixed NMT training recordings (first 300 s, 800-sample Welch segments, i.e. 0.25 Hz bins; $+10^{-20}$ added prior to the log).

The filter is a log10 amplitude gain per channel on that 0.25 Hz grid. It starts at zero, and is later updated four times: shape every subject, re-z-score, take the average of the cohort's log-spectra, add half the remaining gap to NMT's. Shaping multiplies the FFT of the recording by the gain (linearly interpolated onto the FFT's frequency grid without smoothing); the gain is real, so the filter changes no phase (zero-phase). Each recording is reflect-padded by 800 samples at both ends before the FFT, and trimmed after it. The result is z-scored again and divided into 2.5 s frames without overlap.

One filter per channel is fitted on all 15 subjects pooled, with no labels, and the same filter is applied to every subject (Experiment 8); it is refitted in each LOSO fold on the 14 training subjects (Table 19). Four gates were fixed in POSTHOC.md and reported whether or not they passed: mean log-spectrum of the matched cohort within 0.25 decades of NMT's at all channels and bins between 0.5 and 99 Hz; GTJ's input BatchNorm $|z|$ at most 1.5 in every 1 Hz bin, against the running statistics of each NMT checkpoint; embedding spread of GTJ across subjects at least half its spread on 15 NMT evaluation recordings; and at most 5% of its embedding dimensions constant ($\text{std} < 10^{-5}$). All four passed (§5.3.3).

11.9 Data Used by Each Experiment

Table 25: What recordings did each experiment's encoder use to be pre-trained on, what recordings did the trained classifier use, and what recordings did the encoder score. There is no pre-training nor probe training with evaluation recording.

Experiment	Encoder pre-training	Classifier training	Evaluation
1, 2, 4	Released ViT-M: the authors' own pre-training; random-weight twins: none	TUAB, the five 546-recording training subsets and (Experiment 4) the full 2,717-recording training set; Experiment 2 fine-tunes the whole network on the same subsets	TUAB official eval set (276)

Experiment	Encoder pre-training	Classifier training	Evaluation
3	Released ViT-M and its random-weight twins	None (no classifier)	TUAB recordings and the 15-subject stroke cohort (Table 9)
5	FEI, GTJ: the NMT training folds of the seed-0 5-fold split ($\approx 1,934$ recordings), labels unused	The same NMT training fold, with labels	The held-out NMT fold (≈ 483 recordings)
6, FEI row	The NMT-only FEI fold encoders of Experiment 5's three pre-training runs, plus two further FEI-only runs	As Experiment 4 (TUAB training subsets)	TUAB official eval set (276)
6, GTJ and FEI+GTJ rows	NMT training fold plus TUAB's 2,717 unlabelled training recordings (three pre-training runs)	As Experiment 4	TUAB official eval set (276)
7	None (BSI and spectral features)	LOSO: the other 14 stroke-cohort subjects	The held-out subject, all 15 in turn
8	The NMT-only fold encoders of Experiment 5 (all three pre-training runs); released ViT-M	LOSO: the other 14 subjects, after the spectral-matching filter (§11.8)	The held-out subject; also $100\times$ repeated 5-fold and leave-pair-out